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Lesson Notes Part I

MODULE 1 | LESSON 1 | PART I

SAMPLE MOMENTS AND PORTFOLIO PERFORMANCE - PART I

Reading Time	60 minutes
Prior Knowledge	Basic statistics and a working knowledge of basic portfolio theory
Keywords	Return and variance of a portfolio, Beta of a portfolio and relevant types of risk, Performance appraisal (Sharpe and Treynor ratios, Sortino ratio, information and appraisal ratio, skewness, and kurtosis)

This lesson introduces financial portfolios and discusses how to compute certain moments like mean and standard deviation as well as a measure of systematic risk, like the beta. Then, after a discussion of risk, we look at how to assess the performance of portfolios in regards to particular risks. At this stage, the traditional indices used for performance appraisal are introduced, like the Sharpe and Treynor ratios. Furthermore, we discuss how volatility is bad only when it is on the downside and not on the upside, and we also consider other measures of appraisal. Finally, we discuss the implications of skewness and kurtosis and what impact they could have on performance if assessed without regard to these statistics.

1. Sample Moments and Relevant Vector Notation

An investment portfolio is defined as a collection of a variety of assets (stocks, bonds, money market accounts, exchange-traded funds, closed-end funds, real estate, private investments, etc.). Some of these assets were mentioned individually in previous courses. This lesson is not about the specifics of any individual assets; instead, it is focused on an aggregate class of investments. We will learn how to describe portfolios' returns and their risk potential. In this module and the following modules, we will see how to create optimal portfolios and some of the nuances and weaknesses that come with it.

To work effectively with portfolios of financial assets, we must begin with a formal definition. A portfolio consisting of n assets, will be represented by a vector of weights $w = (w_1, w_2, \ldots, w_n)$.

The weights are expressed as the percentage of total wealth allocated to each particular asset. If

TOP

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all weights are non-negative, and they sum up to 1. Whenever short-selling is allowed, it will be mentioned explicitly.

In working with portfolios, depending on the specific applications, it might be necessary to use the language of vectors and matrices. When mathematically convenient, we will use this language to compute certain quantities for our portfolios.

To this purpose, we introduce some basic notation. If x is an $n \times 1$ vector, we will write it as the column vector:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

and its transpose x^T will denote the $1 \times n$ row vector

$$x^T = (x_1 \quad x_2 \quad \dots \quad x_n)$$

Furthermore, by the $n \times m$ matrix X, we mean that

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1m} \\ x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nm} \end{pmatrix}$$

Then, if $r^T = (r_1, r_2, \dots, r_n)$ represents the vector of returns for each of the n assets in a portfolio, and

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_{nn} \end{pmatrix}$$

is the variance covariance matrix of the n-assets, we can write the return for the portfolio as

$$r_p = \sum_{i=1}^{n} w_i r_i = w^T r \text{ and } \sigma_p^2 = \sum_{i=i}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij} = w^T \Sigma w.$$

2. Portfolio Beta

The beta of a security or a portfolio is a measure of the volatility or systematic risk of this security or portfolio compared to the market as whole. The required rate of return on an asset can be related to its systematic and non-systematic risk by writing the capital asset pricing model (CAPM) for the asset a as:

$$E[r_a] = r_{RF} + \beta(E[r_m] - r_{RF})$$

TOP

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where r_m represents the market return, r_{RF} is the risk-free rate, which we assume to be constant over the reference period, and ϵ is an error term such that $E[\epsilon] = 0$ represents the asset's risk not related to market conditions. This model can be re-written as:

$$r_a - r_{RF} = \beta(r_m - r_{RF}) + \epsilon$$

Then, a simple application of results from linear regression yields

$$\beta = \frac{cov(r_a - r_{RF}, r_m - r_{RF})}{var(r_m - r_{RF})},$$

where var and cov are short forms for variance and covariance, respectively.

The beta of a portfolio, say β_p, can be easily obtained from the betas of the individual assets that make up the portfolio. That is, if we represent the beta of each of the n assets as a column vector, i.e.:

$$\beta = (\beta_1\ \beta_2\ \dots\ \beta_n)$$

and w is the vector of the assets' weights in the portfolio, then the formula for β_p is simply:

$$\beta_p = w^T\beta = \sum_{i=1}^{n} w_i\beta_i.$$

To check that the beta of a portfolio indeed has a linear structure, it is sufficient to recognize that, according to the definition of beta we gave above, it follows that

$$\hat{\beta}_p = \frac{cov(r_p - r_{RF}, r_m - r_{RF})}{var(r_m - r_{RF})}.$$

However, we saw earlier that $r_p = \sum_{i=1}^{n} w_i r_i$ and that $\sum_{i=1}^{n} w_i = 1$ and therefore:

$$\beta_p = \frac{cov(\sum_{i=1}^{n} w_i(r_i - r_{RF}), r_m - r_{RF})}{var(r_m - r_{RF})}$$

$$\beta_p = \frac{\sum_{i=1}^{n} cov(w_i(r_i - r_{RF}, r_m - r_{RF}))}{var(r_m - r_{RF})}$$

$$\beta_p = \sum_{i=1}^{n} w_i \frac{cov(r_i - r_{RF}, r_m - r_{RF})}{var(r_m - r_{RF})}$$

$$\beta_p = \sum_{i=1}^{n} w_i\beta_i$$

using the definition of β_i $\textbf{for each}\ i = 1, 2, \dots, n$ as claimed. From this derivation, it should be clear that the beta of a portfolio is a linear combination of the betas of its assets through the li y (bi-linearity, actually) and the other properties of the covariance operator.

TOP

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portfolio is indicated in the table immediately below:

	Asset 1	Asset 2	Asset 3
beta	0.5	1.6	1.2
$ amount	$250,000	$350,000	$400,000

What is the beta for this portfolio?

The dollar value of our portfolio is

$\$250,000 + \$350,000 + \$400,000 = \$1,000,0000$.

Then, the weights for each of the assets in the portfolio can be computed as

$$w_1 = \frac{250,000}{1,000,000} = 0.25, w_2 = \frac{350,000}{1,000,000} = 0.35, w_3 = \frac{400,000}{1,000,000} = 0.4.$$

Thus, the beta for our portfolio turns out to be

$$\beta_p = \sum_{i=1}^{3} w_i \beta_i = 0.25 \times 0.5 + 0.35 \times 1.6 + 0.4 \times 1.2 = 1.165.$$

You will learn about risk management in a different module, but we can still use this example to illustrate the use of the betas of assets.

Suppose that the portfolio manager is concerned that the overall market is going to be bumpy during the next 12 months and, as such, would like to eliminate the systematic risk in their portfolio.

The S&P 500 Index is currently priced at 1,000 and a futures contract on the S&P 500 index is currently priced at $1,010 and has a beta of 1.05. Each future contract is $250 times the index price.

The manager does not want to sell the assets because they think the assets are good long-term investments.

How should the manager act to reduce their market exposure during this period?

Reducing market exposure is equivalent to reducing the beta. More specifically, if we create a new portfolio with a beta of zero, we would have created a **beta neutral portfolio**. For this, the manager can use the S&P 500 futures and sell short:

$$n = \left(\frac{\beta_T - \beta_p}{\beta_F}\right) \times \frac{MVP}{F}$$

with β_T the new target beta, i.e., zero in our case, MVP the market value of our portfolio, and F the market value of the futures contract on the S&P500. Then,

$$n = \left(\frac{0–1.165}{1.05}\right) \times \frac{1,000,000}{250 \times 1010} = -4.394.$$

TOP

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3. Relevant Types of Risk

It is not sensible to talk about portfolio returns without talking about risk because investment decisions involve a trade-off between return and risk. Investors need to be aware of the risk they are assuming and what kind of return they are getting for taking these risks. This can also be useful when deciding which portfolio managers to hire or fire.

One can identify several types of risk:

- **interest rate risk**, caused by changes in the level of interest rates and which generally affect securities inversely, although it affects bonds more directly;
- **market risk**, due to fluctuations in the overall market and to which all securities are exposed, although common stocks are the most sensitive to it;
- **inflation risk**, which affects the purchasing power of investments. This risk is related to interest rate risk as interest rates usually rise with inflation;
- **business risk**, or the risk of doing business in a particular industry;
- **exchange rate risk**, the consequence of currency fluctuations and concerning portfolios that contain global investments since currency fluctuations can affect the variability of returns;
- **liquidity risk**, which is associated in particular with investments that trade in secondary markets. If an investment cannot be bought or sold quickly because it is not liquid, then there could be "price slippage" between when the decision to buy or sell was made and when the transaction actually took place. This is also seen in primary markets when the size of certain orders is large compared to the volume of trades for that particular security; and
- **country risk or political risk** is relevant to investors investing internationally, due to instability at both a political and economic level.

All these categories are part of what we refer to as total risk. However, it is convenient to categorize sources of risk according to their nature, i.e., systematic risk and non-systematic risk. The key "equation" here is

$$\textbf{total risk} = \textbf{systematic risk} + \textbf{non-systematic risk.}$$

Systematic (or market) risk is the risk attributable to broad macro factors and affects all securities. When an investor constructs a well-diversified portfolio, they can eliminate part of the total risk, and what is left is the non-diversifiable portion of the risk, which is known as systematic risk.

Non-systematic (idiosyncratic) risk represents the variability in total returns of an investment that is not related to overall market variability. This type of risk is unique to a particular security and, among the risks listed above, related to liquidity risk, business risk, and country risk.

We have already learned how to "measure" total risk and systematic risk. Next, we will provide a measure of non-systematic risk, which we will be referring to as σ_u. To compute σ_u, one begins with a version of the one-factor model:

$$r_p = E[r_p] + \beta_p(r_m - E[r_m]) + u.$$

TOP

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β_p is a coefficient that measures the portfolio's exposure to the market return r_m, and u is the non-systematic component of the portfolio's return.

The model assumes that r_p and r_m are jointly normally distributed and r_m and u are uncorrelated. Then, using basic results for simple linear regression, we find that

$$\beta_p = \frac{cov(r_p, r_m)}{var(r_m)}.$$

In this one-factor model, the two parts on the right-hand side are uncorrelated with one another and therefore:

$$\sigma_p^2 = \beta_p^2 \sigma_m^2 + \sigma_u^2.$$

From the last equation, we can also write.

$$\sigma_u^2 = \sigma_p^2 - \beta^2 \sigma_m^2$$

and using the value for β_p in this equation, we get, using simple algebraic manipulations, that

$$\sigma_u^2 = \sigma_p^2 (1 - corr^2(r_p, r_m))$$

and we see that if a portfolio has a perfect correlation with the market portfolio, then $\sigma_u^2 = 0$, as we should expect.

4. Performance Appraisal: Sharpe and Treynor Ratios

Managers and investors alike try to appraise the performance of their investments to assess whether investment results are satisfactory. Risk-adjusted performance measures are desirable because it is relatively easy to get lucky by assuming a lot of risk. However, this exposes the investors to potentially serious downsides as well whenever things don't work out as hoped. This is even more important when the wealth of an individual or organization is managed by a third party, like a portfolio manager who is allowed to pursue an active management style in deciding the composition of portfolios for clients. It is important for investors or mutual funds that a portfolio manager's active style adds value on a risk-adjusted basis.

We are now going to look at several appraisal measures, the type of risk they address, and some of the issues.

The notorious Sharpe ratio is calculated as the excess return over the risk-free rate divided by the standard deviation:

$$\text{SR}_p = \frac{\bar{r}_p - \bar{r}_{RF}}{\sigma_p}.$$

where $\bar{r}_p$ and $\bar{r}_{RF}$ denote the mean return on the portfolio and the mean risk-free rate during the r[illegible]ce period, respectively, and σ_p represents the volatility of the portfolio during the same p[illegible] Overall, a Sharpe ratio of less than 1 is considered bad; between 1–1.99, it is considered

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The Treynor ratio is similar to the Sharpe ratio, but the denominator only considers systematic risk rather than total risk, i.e.:

$$T_p = \frac{\bar{r}_p - \bar{r}_{RF}}{\beta_p}.$$

When is the Treynor ratio more appropriate than the Sharpe ratio? Since the denominator in the definition of the Treynor ratio is the beta of that portfolio, it is more appropriate when one can assume efficient markets; that is, the Treynor ratio is a suitable form of portfolio appraisal only in those instances where portfolios just have systematic risk.

The **Modigliani and Modigliani measure** (M2, M^2, RAP, Modigliani risk-adjusted performance) allows us to expand the applicability of the Sharpe ratio. The formula for it is:

$$M2_p = \bar{r}_{RF} + \left(\frac{\bar{R}_p - \bar{r}_{rf}}{\sigma_p}\right)\sigma_m = \bar{r}_{RF} + \mathrm{SR}_p\sigma_m.$$

In essence, M2 is a scaled version of the Sharpe ratio and tells us how much a portfolio would have rewarded an investor with the given amount of risk taken in terms of the risk-free rate of return and benchmark portfolio. In other words, it could help us determine how much return the portfolio would have attained if it had the same risk level and benchmark. Note that if $\sigma_p < \sigma_m$, then the expected return on the portfolio would have been magnified while the opposite is true if $\sigma_p > \sigma_m$.

The M2 overcomes the difficulties associated with the Sharpe ratio and the difficulties with comparing Sharpe, Treynor, Sortino, and other ratios among different investments.

Because M2 yields a risk-adjusted performance measure as a percentage return unit, it is very straightforward to interpret. However, one should notice that the M2 measure is derived directly from the Sharpe ratio, and therefore, any portfolio orderings using the M2 measure will be the same as the portfolio ordering using the Sharpe ratio.

These lesson notes are divided into two parts. Please continue reading about the Sortino Ratio next.

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Lesson Notes Part II

MODULE 1 | LESSON 1 | PART II

SAMPLE MOMENTS AND PORTFOLIO PERFORMANCE - PART II

Reading Time	60 minutes
Prior Knowledge	Basic statistics and a working knowledge of basic portfolio theory
Keywords	Return and variance of a portfolio, Beta of a portfolio and relevant types of risk, Performance appraisal (Sharpe and Treynor ratios, Sortino ratio, information and appraisal ratio, skewness, and kurtosis)

This lesson introduces financial portfolios and discusses how to compute certain moments like mean and standard deviation as well as a measure of systematic risk, like the beta. Then, after a discussion of risk, we look at how to assess the performance of portfolios in regards to particular risks. At this stage, the traditional indices used for performance appraisal are introduced, like the Sharpe and Treynor ratios. Furthermore, we discuss how volatility is bad only when it is on the downside and not on the upside, and we also consider other measures of appraisal. Finally, we discuss the implications of skewness and kurtosis and what impact they could have on performance if assessed without regard to these statistics.

5. Skewness: The Sortino Ratio

Traditional investing theory assumes that investment returns form a symmetrical, bell-shaped curve, with most of the returns in the middle, and large gains and losses falling at the “tails” of the distribution. In reality, return distributions are often tilted a bit—and sometimes more than a bit to the left. This increases the likelihood of “disappointing returns” that are lower than expectations. If one assumes a normal distribution, this assumption fosters a sense of confidence that the portfolio's returns have a low probability of falling more than three standard deviations from the mean, less than 0.15% as a matter of fact. Also, under the assumption of normal returns, the Sharpe ratio provides a reliable appraisal measure to assess the performance of a portfolio. However, the empirical evidence shows that several classes of assets exhibit negative skewness, and when this negative skewness is not negligible, this can become a significant problem.

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portfolios when skewness of returns is taken into consideration.

Regarding the first problem, we notice that the Sharpe ratio does not differentiate between good and bad volatility and does not take skewness into consideration. Most, if not all, investors would not object when the returns are volatile above a certain threshold on the upside. Then, again, most if not all would feel very uncomfortable when volatility hits their portfolios on the downside. Essentially, one drawback of the Sharpe ratio is that the portfolio manager is penalized for all volatility, good and bad, and does not differentiate between the "good" and the "bad" tails.

The Sortino ratio considers the standard deviation on the downside, i.e.

$$\mathrm{SoR}_p = \frac{\bar{r}_p - r_{MAR}}{\sigma_{downside}}.$$

In this case, a target rate of return, MAR (minimum accepted return), is identified and $\sigma_{downside}$ refers to the target semi-standard deviation or the volatility of only those returns that are below the MAR. The Sortino ratio is more appropriate for investments that do not exhibit returns with symmetrical distributions. In this case, using Sharpe ratio, positively and negatively skewed returns would both produce a lower Sharpe ratio. However, only investments with negatively skewed return distributions would result in a lower Sortino ratio. While the Sortino ratio is convincing at dealing with the problem of positive and skewed volatility, the target return is somewhat arbitrary, unlike the risk-free rate, and different investors could define it differently, ending up with different calculated Sortino ratios.

There is also an M^2 version of the Sortino ratio.:

$$M^2_{p,Sortino} = r_p + \mathrm{SoR}(\sigma_{\mathrm{Dbench}} - \sigma_D)$$

where r_p is the annualized portfolio return, σ_{Dbench} is the benchmark annualized downside risk and the σ_D the portfolio annualized downside risk.

Regarding the second difficulty that we raised, that is how to take skewness into consideration when determining the optimal portfolio: One has to introduce the notion of co-skewness. This will be done in the next module. Here, we introduce the definition of skewness, which, unlike how variance or standard deviation involves the second moments of distributions, involves the third moment of distributions.

Skewness is a measure of the asymmetry of a distribution. Negative skewness occurs when the values to the left of the mean are fewer but further from it than the values to the right. Positive skewness occurs if the opposite is true. Skewness (Sk) for a collection of returns $r_1, r_2, \ldots, r_N$ is calculated as

$$\widehat{\mathrm{Sk}} \equiv \hat{\gamma} = \frac{\sum_{i=1}^{N}(r_i - \bar{r})^3}{(N-1)s^3}$$

For samples of points that are not very large, the following modified formula has better statistical properties:

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In both cases, $\bar{r}$ is the sample mean and s the sample volatility of the returns. For large values of n, the two estimates are almost identical.

5.1. Example: The Skew-Normal Distribution

We say that the random variable X has the skew-normal distribution, $X \sim SN(\mu, \sigma, \alpha)$ where μ is the location parameter, $\sigma > 0$ is the scale parameter, and α is the shape parameter, if its probability density function has the following form:

$$f(x) = \frac{2}{\sigma}\phi\left(\frac{x-\mu}{\sigma}\right)\Phi\left(\alpha\left(\frac{x-\mu}{\sigma}\right)\right)$$

with $\phi(\cdot)$ and $\Phi(\cdot)$ being the probability density function (PDF) and the cumulative distribution function (CDF) of a standard normal random variable, respectively.

Note that, unlike in the case of a $N(\mu, \sigma^2)$ random variable, μ and σ are not the mean and the standard deviation of the distribution, except when $\alpha = 0$. As a matter of fact, if $X \sim SN(\mu, \sigma, \alpha)$, then

$$E[X] = \mu + \sigma\delta\sqrt{\frac{2}{\pi}}$$

and

$$\text{variance}[X] = \sigma^2 \cdot \left(1 - \frac{2\delta^2}{\pi}\right)$$

with

$$\delta = \frac{\alpha}{\sqrt{1+\alpha^2}}.$$

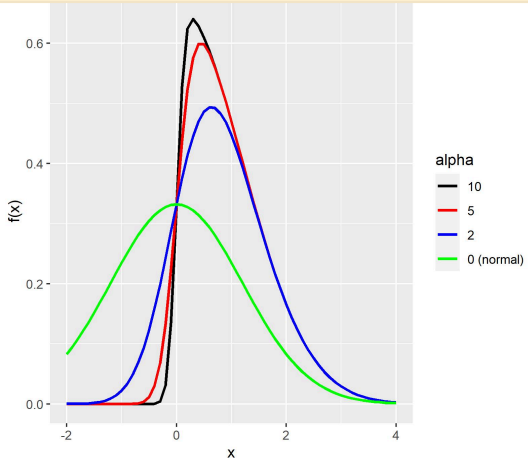
When $\alpha < 0$, the distribution is negatively skewed, and it is positively skewed if $\alpha > 0$. The formula for the skewness is somewhat complicated, but it is available:

or, in terms of just α

$$\gamma_1 = \frac{4-\pi}{2}\left(\frac{\sqrt{2}\alpha}{\sqrt{\pi(1+\alpha^2) - 2\alpha^2}}\right)^3.$$

Furthermore, the rate of decay to 0 of the probability density function as $|x| \to \infty$ is either the same as that of the normal density or even faster, depending on whether α and x have equal or opposite signs. This means that the skew-normal retains normal-like tails in the direction of the skew. A consequence of this fact is that a skew-normal is useful when the distribution of returns has no more outliers than the normal distribution. For this very reason, the skew-normal distribution is unsuitable in applications where the observed data exhibits heavier tails than the normal distribution. This means that when dealing with financial portfolios, the skew-normal family can address problems of skewness but not those of kurtosis (which we will be looking at below).

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One can observe how the tails of skew-normal distributions behave using the following animation (change the parameters to see the changes in real time):

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The ability to use a skew-normal distribution to fit the returns of several stocks is weak (change the parameters to see the changes in real time):

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Maximum likelihood estimates for μ, σ and α can be computed numerically, although no closed-form expression is available. For an independent sample $x_1, x_2, \ldots, x_n$ the log-likelihood function is given by:

$$l(\mu, \sigma, \alpha; x_1, \ldots, x_n) = \text{ constant } - n \ln(\sigma) - \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{2\sigma^2} + \sum_{i=1}^{n} \xi_o\left(\frac{\alpha(x_i - \mu)}{\sigma}\right),$$

with

$$\xi_0(x) = \ln(2 \cdot \Phi(x)).$$

To complete this section, we'll look at one more aspect. When we consider a sequence of returns for an investment or a portfolio, we don't work with the population of all possible returns, but we have a sample of them available. If we compute the skewness using the formula given above, the result could change with the particular sample used. In other words, the sample skewness is a random variable. The practical implication is that we should test whether it is zero or not. This can be done easily:

$$H_0 : \gamma = 0 \quad \text{vs.} \quad H_1 : \gamma \neq 0.$$

Using $\hat{\gamma}$ obtained from the sample, if the number of observations N is large and under the assumption of normality, the distribution of the sample skewness is itself normal with mean 0 and standard error $\text{S.E.}(\hat{\gamma}) = \sqrt{6/N}$. This means that we can test whether the skewness is zero or not using the following test-statistics:

$$z = \frac{\hat{\gamma} - 0}{\sqrt{6/N}}$$

which has a standard normal distribution.

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(Students may like to peruse the following paper by Rui Albuquerque: "Skewness in Stock Returns: Reconciling the Evidence on Firm versus Aggregate Returns")

6. Information and Appraisal Ratio

The **information ratio** (IR) is used to measure the performance of a portfolio against a benchmark while accounting for the differences in overall risk. It is defined as:

$$IR_p = \frac{\bar{r}_p - \bar{r}_b}{\sigma(r_p - r_b)},$$

with $\bar{r}_p, \bar{r}_b$ representing the mean return for the portfolio and the portfolio benchmark, respectively. In this case, the denominator $\sigma(r_p - r_b)$ can be interpreted as the **tracking risk** or the volatility of the portfolio returns vs. that of its benchmark. This characteristic makes the information ratio particularly suitable for comparing the performance of portfolio managers who were assigned pre-defined benchmarks.

The appraisal ratio is a form of signal-to-noise ratio used to measure a fund or portfolio's performance. It is calculated by taking the alpha of the investment of the portfolio and dividing it by its non-systematic risk. The alpha represents excess return, calculated as the return earned by the portfolio over the return implied by the CAPM model. Its formula is $AR_p = \frac{\alpha}{\sigma_u}$.

Now, the alpha is measured by comparing the return in excess of that required by the CAPM, i.e.,

$$\alpha = r_p - (r_{RF} + \beta_p(r_m - r_{RF}))$$

and
σ_u is as it was derived in part 3 of this lesson, i.e.: $\sigma_u = \sigma^2_{r_p}(1 - corr^2(r_p, r_m))$.

7. Kurtosis and the Problem of Heavy Tails of Distribution

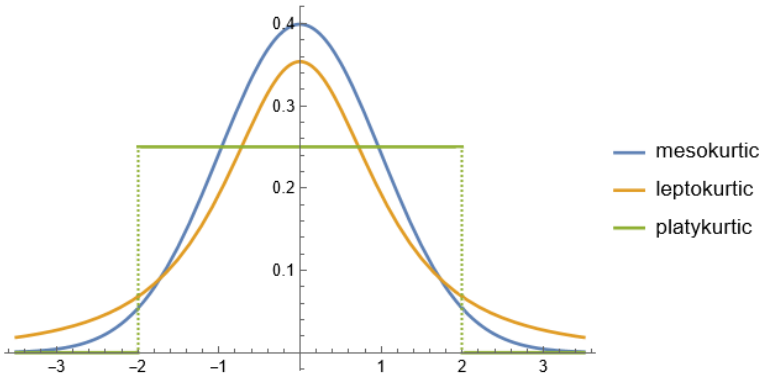
In statistics, the term kurtosis is used to describe the distribution of the dataset, and that makes it possible to measure to what extent the data points of a particular distribution differ from the data of a normal distribution. We are aware that for a normal distribution, less than 0.15% of its values fall above or below three standard deviations from the mean. However, if a distribution has heavier tails than the normal, it could have more extreme values. Kurtosis can also be combined with skewness.

In finance, kurtosis is used to measure the amount of financial risk associated with any instrument or transaction. The more kurtosis, the higher the financial risk associated with the concerned dataset. We could say that, while skewness is a measure of symmetry in distribution, kurtosis is the measure of heaviness of the tails of distributions. We distinguish distribution functions according to which of the three types of kurtosis they exhibit:

- mesokurtic distribution if the distribution has a kurtosis statistic similar to that of the normal distribution;
- leptokurtic distributions if the tails of the distributions are longer than those for the normal distributions, i.e., the population has more outliers than the normal distribution; and

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Figure 2: Different Types of Kurtosis



The formula for the kurtosis (K) of a random variable X is given by:

$$\mathrm{K}(X) = \left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = \frac{\mu_4}{\sigma^4}$$

which tells us that kurtosis is the fourth standardized moment of the distribution of X with μ_4 the fourth central moment and σ the standard deviation.

The theoretical formula used in the definition above is good, but the required moments μ_4 and σ are seldom known, and one needs to use the sample kurtosis, which for a sample of size N, $X_1, X_2, \ldots, X_n$ is defined as

$$\widehat{\mathrm{K}}(X) = N \cdot \frac{\sum_{i=1}^{N}(X_i - \bar{X})^4}{(\sum_{i=1}^{N}(X_i - \bar{X})^2)^2}.$$

There is another formula with better properties for estimating kurtosis:

$$\widetilde{\mathrm{K}}(X) = \frac{N(N+1)(N-1)}{((N-2)(N-3)} \cdot \frac{\sum_{i=1}^{N}(X_i - \bar{X})^4}{(\sum_{i=1}^{N}(X_i - \bar{X})^2)^2}.$$

Again, for large values of N, the two formulae yield very similar estimates.

The normal distribution has a kurtosis $= 3$. Then, we could say that a distribution is platykurtic if its kurtosis is < 3 and leptokurtic if its kurtosis is > 3, instead. Sometimes, out of convenience, we use the **excess kurtosis**, which is simply kurtosis -3. In this way, we can talk of mesokurtic distributions if their excess kurtosis is near 0, platykurtic when the distribution has negative excess kurtosis, and leptokurtic when the distribution has positive excess kurtosis.
The uniform distribution has an excess kurtosis of -1.2, and the logistic distribution has an excess kurtosis of 1.2 instead.
The t-distribution with ν degrees of freedom has an excess kurtosis $\frac{6}{\nu-4}$ for $\nu > 4$, infinite for $\nu = 3, 4$, and undefined for $\nu = 1, 2$.

The binomial distribution with parameters n and p is found to have an excess kurtosis equal to $\frac{1-6p(1-p)}{np(1-p)}$, which can go from positive to negative depending on the value of the parameter p.
In finance, the main problem associated with the "heaviness" of tails is that using the normal

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7.1. Example: Student's t-distribution

A probability often used to represent situations where the distribution of returns has heavier tails than a normal one is Student's t-distibution. The functional form of its PDF is

$$f(x) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2}.$$

where $\nu > 2$ is a parameter that represents the degrees of freedom. We write $t_{(\nu)}$ or $t(\nu)$ to denote this distribution. It has mean 0 and its variance is equal to $\frac{\nu}{\nu-2}$.

a$t_{(\nu)}$ distribution is symmetrical about the mean (in our case 0 although there are versions of Student's t also for non-zero means, non-central t-distribution) like the normal distributions, and therefore it has zero skewness. However, it has heavier tails than the normal distribution has.

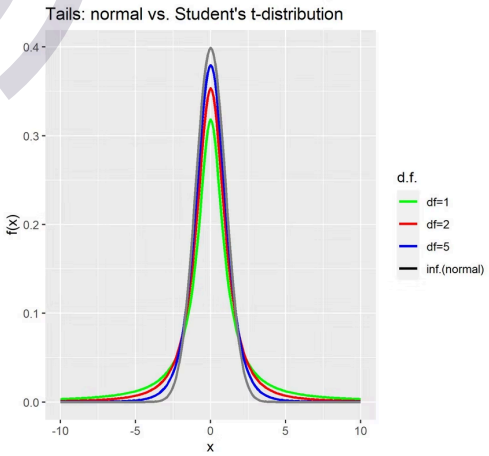
Another important feature is that as the number of degrees of freedom increases towards ∞, the distribution becomes closer to a normal distribution. In other words, the standard normal distribution can be considered the limit of a Student's t-distribution as the degrees of freedom increase to infinity. For most practical purposes, the approximation is already strong even for $\nu = 100$ and perhaps even just $\nu = 60$.

Like for the normal, if it is desired to retain certain features of the Student's t-distribution in terms of heaviness of the tails, but we want to introduce some skewness in our applications, there is a version of this distribution called the skewed t-distribution. It is actually easy to define if practitioners need it for their work:

$$f(x, \nu, \alpha) = 2 \times f(x, \nu) \times T(\alpha \cdot x, \nu),$$

with ν the degrees of freedom, $T(\cdot, \nu)$ the distribution function of a Student's t-distribution with ν degrees of freedom, and α a shape parameter that introduces a slant in the distribution.

Figure 3: Normal vs. Student's t-distribution Tails



As we did in the case of the sample skewness, we are interested in testing whether the distribution of, say, a specific sequence of returns exhibits fat tails or kurtosis. We would like to perform the test:

$$H_0 : k = 0 \quad \text{vs.} \quad H_1 : k \neq 0,$$

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$$z = \frac{k - 0}{\sqrt{24/N}}$$

which has a standard normal distribution.

It is possible to test for normality by combining the test for skewness and kurtosis into a single test. In this case, the test is known as the Jarque-Bera test:

$$\text{JB } = N\left(\frac{\hat{\gamma}^2}{6} + \frac{(\hat{k} - 3)^2}{24}\right)$$

and which has a chi-square distribution with 2 degrees of freedom.

The skewness-kurtosis ratio is the division of skewness by kurtosis.

$$\text{Skewness-Kurtosis ratio} = \frac{\text{Skewness}}{\text{Kurtosis}}$$

It is used in conjunction with the Sharpe ratio to rank portfolios. The higher the rate, the better. In fact, since kurtosis is always positive, the ratio being bigger means that there is more "good tail" in our distribution of returns.

There exists also an adjusted Sharpe ratio (introduced by Pezier and White in 2006) to adjust for skewness and kurtosis by incorporating a penalty factor for negative skewness and excess kurtosis:

$$\text{adjusted-Sharpe ratio} = SR \cdot \left[1 + \frac{Sk}{6} \times SR - \frac{K - 3}{24} \cdot SR^2\right]$$

where SR is the Sharpe ratio (using annualized data), Sk the skewness, and K the kurtosis of returns.

The performance measures presented in this lesson are just a sample of many, although they are among the best known and most used. The lesson one should take away is that there are many ways to assess portfolio performance, and each has its strengths and weaknesses. But when used in conjunction, and not blindly, they can often provide a more complete picture.

8. Conclusion

We have gained an appreciation for sample moments and betas and how they can help to appraise the performance of a portfolio. We have also seen that skewness and kurtosis of returns create problems; in particular, the behavior of the tails of the probability distribution of returns can make an analysis based on bell-shaped distributions less valid or even misleading. In the next lesson, we will look at how to measure the risk on the tails of return distributions when we look at Value at Risk (VaR).

References

- Albuquerque, Rui. "Skewness in Stock Returns: Reconciling the Evidence on Firm Versus Aggregate Returns ." *Review of Financial Studies*, vol. 25, 2011, pp. 1–56.

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Lesson Notes

MODULE 1 | LESSON 2

VALUE AT RISK (VaR)

Reading Time	120 minutes
Prior Knowledge	Ability to read in from datasets from files or via data feeds using software, some understanding of statistical distributions, notion of simulations, some coding ability to make concepts operational.
Keywords	Value at Risk (VaR) and Expected Shortfall (ES, or cVaR), methods to calculate VaR, Monte Carlo simulations, VaR with non-stationary distributions, main criticisms of VaR, going from 1-day VaR to n-day VAR

In the previous lesson, we looked at important statistics about the return of portfolios and learned how to assess performance. In this lesson, we will examine how to measure the riskiness of portfolios or investments. This is not as simple as it might look because one has to deal with problems stemming from the unknown distribution of returns or, worse, the fact that the distribution may be non-stationary as financial markets experience changes of regime and/or volatility tends to present itself in clustered form.

1. The Role of Value-at-Risk (VaR)

Portfolio performance does not address one key question: how risky is an investment, or more bluntly, how much could we lose?

One could say that in the worst case scenario we could lose all or 100% of our investment (through either financing or derivatives), while in the best case scenario, we could make a boatload of money. However, this answer is not very useful in practice neither to provide a measure or risk nor to manage it. Instead, it would seem that being able to state something like "our portfolio would not lose more than, say 10% with a probability of 95%" has much more value for a practitioner. This is what Value at Risk (**VaR**) is supposed to accomplish.

So, what is Value at Risk? It refers to the maximum potential loss which shall not be exceeded during a specified period of time with a given probability level.

Our goal in this lesson is to learn as much as possible about several of the nuances that are involved using VaR. In order to make the definition above more operational we need to translate the concept to a more operational format. VaR requires two parameters:

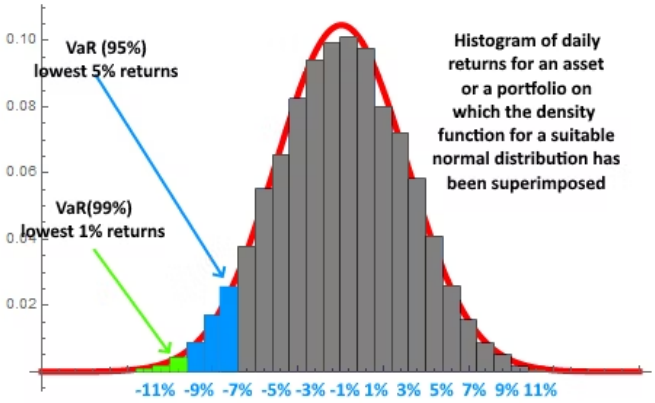
- a confidence level ($1-\alpha$); and
- a time horizon or reference period (T).

Furthermore, we will use $\mathcal{L}$ to denote the (daily) losses over the holding period T and we will use it as $\mathcal{L}=-\mathcal{R}$, with $\mathcal{R}$ representing the (daily) returns on our investment or portfolio. Sometimes, it is easier to communicate in terms of losses instead of returns turned negative. For a portfolio containing short positions, losses would refer to the securities having positive returns.

We use the symbol $\mathrm{VaR}(1-\alpha)$ to denote the $(1-\alpha)$-th percentile, or the α-th upper percentile of $\mathcal{L}$, that is the lower α-th percentile for returns. Therefore, for continuous loss distributions, we can formally define $\mathrm{VaR}(\alpha)$ as the number such that:

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Figure 1: The Concept of VaR via a picture



If we stated that our portfolio could lose $\$10$ millions over a time horizon of one month with confidence level 98% , this would be equivalent to saying that the 1-month $\mathbf{VaR}(1-\alpha=0.98)$ is $10,000,0000$, or that there is a probability of 98% not to lose more than $\$\ 10$ millions over the next month. Equivalently, this also tells us that there is a probability of 2 % to lose more than $\$\ 10$ millions over the next month.

When the joint distribution of returns for several stocks are normally distributed (or more generally speaking, have a joint distribution that belongs to the class of elliptically contoured distributions), because of diversification, the VaR for a portfolio is less than or equal to the sum of the individual VaRs for each of the individual assets in the portfolio. This happens because in this case is determined as $VaR(1-\alpha)=Mq(\alpha)\sigma_p$ where M is the asset under management and $q(\alpha)$ is the suitable percentile for the elliptically contoured distribution. This family included the normal distributions and the t-distributions among many others. We can show this with two assets, A and B and assuming that both assets have zero expected returns. Also, we have M_A invested in asset A, M_B invested in asset B, with $M=M_A+M_B$ the value of our portfolio. Then, if $w_A=M_A/M$ and $w_B=M_B/M$,

$$\begin{aligned}\mathbf{VaR}_p^2 &= M^2q(\alpha)^2(w_A^2\sigma_A^2+w_B^2\sigma_B^2+2w_Aw_B\rho_{AB}\sigma_A\sigma_B)\\ &= (q(\alpha)M_A\sigma_A)^2+(q(\alpha)M_B\sigma_B)^2+2\rho_{AB}(q(\alpha)M_A\sigma_A)(q(\alpha)M_B\sigma_B)\\ &= \mathbf{VaR}_A^2+\mathbf{VaR}_B^2+2\rho_{AB}\mathbf{VaR}_A\cdot\mathbf{VaR}_B\leq(\mathbf{VaR}_A+\mathbf{VaR}_B)^2.\end{aligned}$$

Taking square roots on both sides yields the desired conclusion. In general, for m asset portfolios:

$$\mathbf{VaR}_p\leq\sum_{i=1}^{m}\mathbf{VaR}_i,$$

with equality holding only when the correlations among all assets are equal to 1. In Section 5 we provide one example where VaR fails to be subadditive in general.

It is also possible to express VaR as a relative loss (% loss) and not just as an amount. Some books or papers distinguish these two ways to measure VaR using $\mathbf{VaR}(\alpha)$ for amount of loss and $\overline{\mathbf{VaR}}(\alpha)$ when the loss is measured as a % loss.

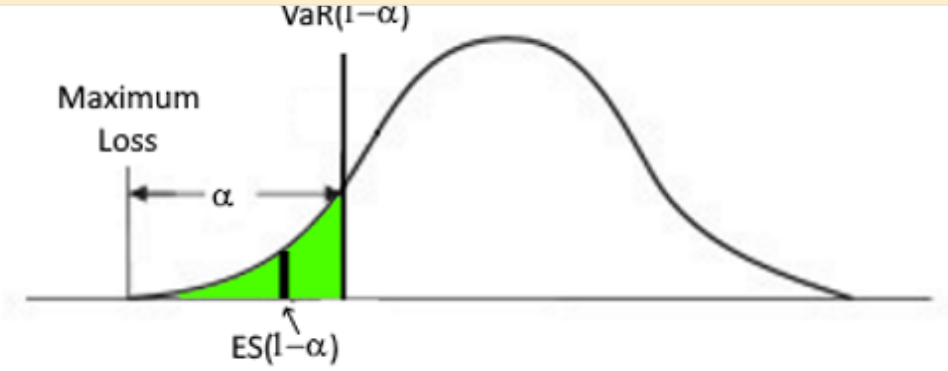
VaR is not completely satisfactory because it primarily tells us that we have a, say 2% probability of losing more than 10 % over a one-month period. However, a portfolio manager is likely concerned about the expected loss whenever the loss exceeds the VaR threshold. To address this concern, we introduce the expected loss given that the loss exceeds the VaR threshold. This measure has been given the technical name of **expected shortfall** (**ES**) (also known as **conditional value at risk** (**cVaR**)) and it is formally defined as:

$$\mathbf{ES}(1-\alpha)=\frac{\int_0^\alpha\mathbf{VaR}(1-s)\,ds}{\alpha}.$$

or, using the language of conditional expectations (and assuming for simplicity that losses have a continuous distribution function),

$$\mathbf{ES}(1-\alpha)=E[\mathcal{L}|\mathcal{L}\geq\mathbf{VaR}(1-\alpha)]$$

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How does one choose a confidence level? Usually, the choice depends on the specific use of $\mathbf{VaR}$. A financial institution monitoring the daily activities of its traders is likely to use a 95% confidence level. However, when dealing with capital requirements, supervisory authorities usually adopt a 99% confidence level. On the other hand, the Basel Committee on Banking Supervision (BCBS) sets the confidence level at a remarkably high 99.9%.

2. Methods to Calculate VaR

There are two main methodologies for VaR calculation:

- **parametric method**, which uses a specific parametric distribution function like a normal distribution or a t-distribution; and
- **non-parametric or historical method**, which uses the empirical distribution obtained directly from the historical data.

It is possible to combine the two methods in what is called a **semi-parametric method**.

Some may also list Monte Carlo simulations as a separate group, but technically, that is just a particular way to handle parametric distributions or even non-parametric ones when simulations are implemented via bootstrapping techniques.

As we are aware, economic and financial time series are usually not stationary. While assuming stationarity may not always be wrong as there are periods in which underlying risk factors are stable, over longer time horizons, that assumption does not hold. As such, we distinguish between two additional types of calculations:

- **unconditional VaR,** based on the unconditional distribution of the underlying risk factors; and
- **conditional VaR**, which assumes that the relevant distribution and/or parameters could change over time.

The non-parametric VaR estimation is simple in principle. Suppose that we want a confidence coefficient of $1-\alpha$ Then all that is required is to estimate the lower $1-\alpha$-percentile of the return distribution, or equivalently the α-percentile of the loss distribution for the sample of historic returns. If we define the losses l_i as the negative of the returns, i.e., $l_i = -r_i$, let the percentile $q(\alpha)$ and its sample equivalent $\hat{q}(\alpha)$ denote the loss that leaves a fraction α of losses above it and the fraction $1-\alpha$ below it.

If our current portfolio is worth M then the nonparametric estimate of VaR is

$$\widehat{VaR}_{np}(1-\alpha) = M \cdot \hat{q}(\alpha).$$

The expected shortfall has a more complicated expression but in reality is a simple concept, i.e.:

$$\widehat{ES}_{np}(1-\alpha) = M \cdot \frac{\sum_{i=1}^{n} l_i \cdot I(l_i > \hat{q}(\alpha))}{\sum_{i=1}^{n} I(l_i > \hat{q}(\alpha))}.$$

The indicator function $I(l_i > \hat{q}(\alpha)) = 1$ if the l_i-loss is more than $\hat{q}(\alpha)$ and 0 otherwise, $\sum_{i=1}^{n} I(l_i > \hat{q}(\alpha))$ counts the number of losses above the threshold $\hat{q}(\alpha)$.

Hence, the formula supplied is just a fancy (but mathematically rigorous) way to write the sample mean of returns that fall below the $\widehat{\mathrm{VaR}}_{np}(1-\alpha)$ threshold.

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We work with the daily returns for AMZN stock during the period from Jan 1, 2015 to Dec 31, 2021. If we wanted to calculate the $\mathrm{VaR}_n p(\alpha)$ for a 1-day period for $\alpha = 0.10, 0.05$, and 0.01 we could use the code below. The small program also produces a histogram of the daily returns and superimposes the suitable normal density. We notice that the histogram exhibits heavier tails than those of the normal distribution. Because of this, the nonparametric VaR could be actually more desirable in this instance. For the sake of simplicity, we assume $M = \$1$.

```
# import required libraries:

import pandas as pd
import numpy as np
from pandas_datareader import data as pdr
import yfinance as yf
yf.pdr_override()
import matplotlib.pyplot as plt
import math
import seaborn
import datetime
from datetime import date

from scipy.stats import norm
from tabulate import tabulate

# read the data for the required stock
ticker ="AMZN"
start_date = datetime.date(2015,1,1)
end_date = datetime.date(2021,12,31)

data = pdr.get_data_yahoo(ticker, start=start_date, end=end_date)['Adj Close']
stock_returns = data.pct_change()
stock_returns= stock_returns.dropna()

VaR_90=np.quantile(stock_returns, .1)
VaR_95=np.quantile(stock_returns, .05)
VaR_99=np.quantile(stock_returns, .01)

table_VaR = [['90%', VaR_90], ['95%', VaR_95], ['99%', VaR_99]]
print(tabulate(table_VaR, headers=['Confidence level', 'VaR']))

ES_90 = stock_returns[stock_returns < VaR_90].mean()
ES_95 = stock_returns[stock_returns < VaR_95].mean()
ES_99 = stock_returns[stock_returns < VaR_99].mean()

table_ES = [['90%', ES_90], ['95%', ES_95], ['99%', ES_99]]
print(tabulate(table_ES, headers=['Confidence level', 'ES']))
```

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```
Confidence level           VaR
------------------  ----------
90%                 -0.0181798
95%                 -0.0279091
99%                 -0.0523702
```

and

```
Confidence level            ES
------------------  ----------
90%                 -0.0319155
95%                 -0.0415431
99%                 -0.0634476
```

We compare now the histogram of the stock's returns over the period with the estimated parametric normal family. The following lines are also taken from the previous program:

```
#pip install pandas-datareader  #Make sure you have at least version 0.10
from scipy.stats import t
mean = np.mean(stock_returns)
stdev = np.std(stock_returns)
nu, mu, sigma = t.fit(stock_returns)
stock_returns.hist(bins=40,density=True,
                       histtype='stepfilled', alpha=0.5)
x=np.linspace(mean-4*stdev, mean + 4*stdev, 100)
plt.plot(x,norm.pdf(x, mean, stdev), "r", label="normal")
plt.plot(x,t.pdf(x, nu,mu, sigma), "g", label="t-dist")
plt.xlabel('returns')
plt.ylabel('frequency')
plt.legend(loc="upper right")
plt.show
```

Figure 3: Comparing the Histogram of Amazon's Daily Returns with the Estimated Normal and t-Distribution

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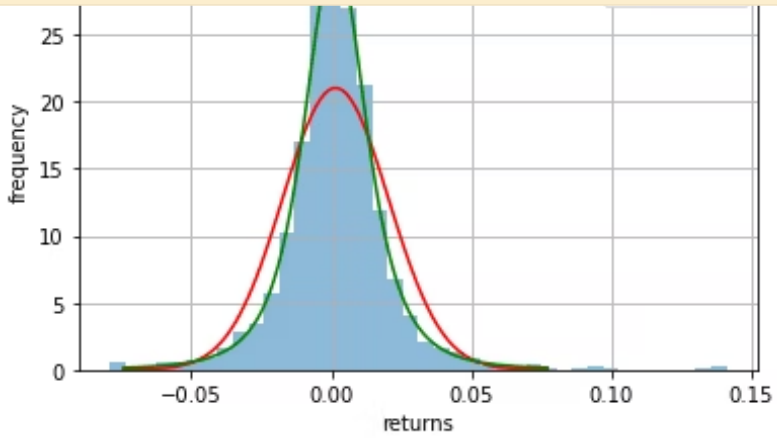


Figure 3 shows that the tails of the histogram of daily returns of AMZN during the reference period are heavier than those for the normal distribution. In this case, using a parametric VaR with a normal distribution could underestimate the Value at Risk, and the t-distribution does a better job at capturing the histogram. So our next step will be to incorporate some of this in our approach to estimating VaR.

Non-parametric estimates using sample quantiles, like in our example, have the advantage that practitioners are not required to make assumptions about the distribution of returns. However, they do work better when the sample size is large and α is not too small. In particular, if α is chosen small, then a very large sample size is required to obtain a reliable estimate of the relevant quantile. The reason is that we have very few data *observations* on the tails compared to the central part of the distribution, so the statistical estimates can lack precision.

When we assume that the stock returns are modeled by a specific parametric distribution, $F(r, \theta = (\theta_1, \ldots, \theta_p))$, all one has to do is obtain an estimate of the parameter(s) $\theta_1, \ldots, \theta_p$ using either MLE (maximum likelihood) or MOM (method of moments) applied to the historical returns available. Then, in general, using a parametric technique, we have

$$\widehat{\text{VaR}}_p(1-\alpha) = M \cdot F^{-1}(\alpha, \hat{\theta}),$$

where M is again the (dollar) value of the asset owned.

Example (continued). Let's say that we are willing to assume that the underlying distribution for the historical returns of AMZN is normal, i.e.:

$$F(r, \theta) = \Phi(\mu, \sigma^2)$$

where $\Phi(\cdot)$ is the cumulative distribution function for a normal family and μ and σ^2 are the two parameters required to describe it.

For a normal distribution, the MLE estimators are $\hat{\mu} = \bar{x}$, the sample mean and $\hat{\sigma}^2 = s^2$, the sample variance (divided by n instead of $n - 1$, but for large samples, that hardly makes any difference). Thus:

```
mean
Out[4]: 0.0015299261202024064
stdev
Out[5]: 0.018958006671941956
pow(stdev, 2)
Out[9]: 0.0003594060169733957
```

i.e., we have found $\hat{\mu} = 0.00153$ and $\hat{\sigma} = 0.01896$.

This means that the parametric model for returns is $r \sim N(\mu, \sigma^2)$, which means also that

$$\frac{r - \mu}{\sigma} \sim N(0, 1),$$

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$$\widehat{\text{VaR}}_p(1-\alpha) = F^{-1}(\alpha, \hat{\theta}) = \hat{\mu} + \hat{\sigma} \cdot q(\alpha).$$

Then, a few lines of Python code yield:

```
import scipy.stats as st
q_010 = st.norm.ppf(.10)
q_005 = st.norm.ppf(.05)
q_001 = st.norm.ppf(.01)

VaR_90 = mean + stdev*q_010
VaR_95 = mean + stdev*q_005
VaR_99 = mean + stdev*q_001

table_VaR = [['90%', VaR_90], ['95%', VaR_95], ['99%', VaR_99]]
print(tabulate(table_VaR,
headers=['Confidence level (normal)', 'VaR']))

Confidence (normal)          VaR
---------------------  ----------
90%                    -0.0227657
95%                    -0.0296532
99%                    -0.042573
```

As we suspected from looking at the figure with the histogram, the $\text{Var}_p(\alpha)$ is lower (bigger losses) than $\text{Var}_{np}(\alpha)$ for $\alpha = 0.10$ and $\alpha = 0.05$. The opposite is true for $\alpha = .01$.

The computation of $\text{ES}(1-\alpha)$ is less straightforward but reasonably simple. If we use a normal distribution, we get

$$\widehat{\text{ES}}_p(1-\alpha) = M \cdot \left(\hat{\mu} - \hat{\sigma}\left(\frac{\phi(\Phi^{-1}(\alpha))}{\alpha}\right)\right),$$

where $\hat{\mu} = \bar{x}, \hat{\sigma} = s$ and $\phi(\cdot), \Phi(\cdot)$ are the pdf and the cdf of a standard normal distribution, respectively.

Here are the few lines of Python code (to be used together with the rest of the code, of course):

```
ES_90 = +mean - stdev*(norm.pdf(q_010)/.10)
ES_95 = +mean - stdev*(norm.pdf(q_005)/.05)
ES_99 = +mean - stdev*(norm.pdf(q_001)/.01)

table_ES = [['90%', ES_90], ['95%', ES_95], ['99%', ES_99]]
print(tabulate(table_ES,
headers=['Confidence level (normal)', 'ES']))

Confidence level (normal)           ES
--------------------------  ----------
90%                         -0.0317411
```

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Following our discussion about the tails of probability distributions and having noticed that the histogram seems to have heavier tails than the normal, we could compute a new parametric VaR using a t-distribution instead of a normal. Say that the chosen parametric family of distribution is a$t(\nu, \mu, \sigma)$, where ν represents the degrees of freedom, μ is the location parameter and σ the scale parameter.

This distribution can be considered the outcome of starting with $X \sim t(\nu)$, a standard t distribution with ν degrees of freedom and considering the new random variable

$$Y = \mu + \sigma X.$$

In this case:

$$\widehat{\text{VaR}}_p(1-\alpha) = M \cdot \left(\hat{\mu} + \hat{\sigma} \cdot F_{\hat{\nu}}^{-1}(\alpha)\right)$$

and

$$\widehat{ES}_p(1-\alpha) = M \cdot \left(\hat{\mu} - \hat{\sigma}\left(\frac{f_{\hat{\nu}}(F_{\hat{\nu}}^{1-}(\alpha))}{\alpha}\right) \cdot \left[\frac{\hat{\nu} + (F_{\hat{\nu}}^{-1}(\alpha))^2}{\hat{\nu} - 1}\right]\right)$$

where $f_\nu(\cdot)$ and $F_\nu(\cdot)$ stand for the pdf and cdf of a standard t-distribution with ν degrees of freedom, respectively and $\hat{\mu}, \hat{\sigma}$, and $\hat{\nu}$ are the estimates (usually MLEs) of the unknown parameters.

To estimate the parameters in Python, we can use the **t.fit** command. However, to run this module, the dataset must be of type array. **t.fit**, by default, uses ML to estimate the parameters, but MOM is also available. Nevertheless, one has to be careful that the t-distribution does not have certain moments for all values of degrees of freedom.

Example (continued). Returning to our case,

```
from scipy.stats import t
from array import array

stock_ret_array = stock_returns.to_numpy()
nu, mu, sigma = t.fit(stock_ret_array)
##############
t.fit(stock_ret_array)
Out[238]: (2.9432547431402076, 0.0014122907770148384, 0.011709608784175964)
```

We found that $\hat{\nu} = 2.94345, \hat{\mu} = 0.001412$, and $\hat{\sigma} = 0.01171$, and these estimates can be used to estimate the required VaR and ES.

```
q_010 = st.t.ppf(.10, nu, mu, sigma)
q_005 = st.t.ppf(.05, nu, mu, sigma)
q_001 = st.t.ppf(.01, nu, mu, sigma)

VaR_90 = q_010
VaR_95 = q_005
VaR_99 = q_001

table_VaR = [['90%', VaR_90], ['95%', VaR_95], ['99%', VaR_99]]
print(tabulate(table_VaR,
```

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```
Confidence level with (t-dist.)           VaR
--------------------------------  ----------
90%                               -0.0178653
95%                               -0.0263626
99%                               -0.0525860
```

3. Monte Carlo Simulations

In principle, this method is very simple and can be seen as an extension of the parametric approach to VaR. In essence, in its most basic form of implementation, we start with a parametric model and simulate, say $n = 1,000,000$ random variables or patterns of prices, over a desired period of time. Then, we can compute the percentile for the required level of confidence and we are done.

Like in any parametric model, the distribution is specified, but the parameters of the same must be estimated using historical data for instance.

Using again the example we were working on previously, here are a few basic lines of code to implement the concept:

Example (continued).

```
n_sims = 1000000
sim_returns = np.random.normal(mean, stdev, n_sims)
sim_VaR = np.percentile(sim_returns, 1)
print('1-day simulated VaR is ', sim_VaR)

1-day simulated 99% VaR is  -0.04257348749817126
```

If a practitioner needed, say a 1-month VaR instead of a 1-day VaR for any level of confidence, all they must do is adjust the parameters to reflect *average return and volatility over a period of 250/12 = 21 days* , but otherwise, the same few lines suffice. However, one needs to be aware that for a 1-month VaR, estimates would be based on fewer observations and thus less accurate.

Of course, there is minimal advantage in using Monte Carlo simulations in such a simplistic way vs. using the parametric or non-parametric methods. Also, if we lack reliable data, the parameters could not be estimated with sufficient precision.

As one has probably guessed already, this is not the end of the story. Monte Carlo methods are very powerful in allowing us to introduce possible scenarios. For instance, let's say that we are interested in a 30-day VaR, but we sense that certain events could take place that could affect the prices and therefore returns of a stock or some portfolio. When using simulations, it is reasonably easy to introduce jumps or model individual risks—something that could not be done using the other two approaches.

So far, we considered the case in which we have one stock. That could very well have been a portfolio considered as an aggregate. Where the approach shows its strength is in handling complex situations involving many assets at the same time because we could simulate each asset separately and consider specific sources of risk for each of the components. However, while in principle it is possible to simulate the returns of each of the, say m assets in the portfolio and then aggregate the returns via a linear combination with known weights, that would not be the most general way to do it.

Assets are not completely uncorrelated with each other and some investment classes may even exhibit strong dependence. This dependence should be taken into consideration. During the Great Financial Crisis, for instance, the world became painfully aware that investments were way more correlated than previously assumed. Monte Carlo simulations can help us model these situations. We will use a multivariate normal approach, but one could use a

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Assume that we have m assets in our portfolio and that these are linearly priced or approximately so.

What do we mean by this? Clearly, if one of the assets is a stock, then its value is a linear function of its price. But, the relationship between a bond and the interest rate is not linear, nor is the relationship between a call option and the price of its underlying stock. However, here we assume either that the relationship is linear or it can be approximated reasonably well by a linear relationship.

In the literature, this is sometimes called a linear portfolio. The situation can be described as follows:

- S is an $m \times 1$ vector of market prices and interest rates;
- Δt represents the time horizon over which we would like to perform risk measurements;
- $V(S,t)$ stands for the value of our portfolio at time t and market prices S;
- $L = V(S,t) - V(S + \Delta S, t + \Delta t)$ is the loss in the value of our portfolio over the time frame Δt ;
- when we say that we have a linear portfolio, we mean that $\Delta V = \beta^T \Delta S$, where β is a vector of sensitivities.

If one of the assets, say W is a derivative, we know from previous courses that

$$\Delta W \approx \frac{\partial W}{\partial t}(\Delta t) + \beta(\Delta S) + \frac{1}{2}\Gamma(\Delta S)^2$$

with $\beta = \frac{\partial W}{\partial S}$, and $\Gamma = \frac{\partial^2 W}{\partial S^2}$, and $\Delta S, \Delta T$ just the discrete time change in value of S and t, respectively. If the first and third terms in the above formula are small, we could approximate the asset in a linear fashion. Incidentally, this tells us that estimating the sensitivities can require some work as well.

To illustrate how the methods work, assume that the relative changes in risk factors (think of returns in case the risk factors are the stock prices) for the next day have a multivariate normal distribution:

$$r_{T+1} \sim N(\mu, \Sigma)$$

where T is the current time, μ an $m \times 1$ vector and $\mathbf{\Sigma}$ is an $m \times m$ variance-covariance matrix, and both need to be estimated. Now, using the assumption of linearity, we have in general that

$$\frac{V_t}{V_{t-1}} = \beta^T r_t$$

which is why we insisted on a linearly priced portfolio. Then, this means that

$$\frac{V_{T+1}}{V_T} \sim N(\beta^T \mu, \beta^T \mathbf{\Sigma} \beta).$$

Now, one gets

$$\widehat{\mathrm{VaR}}(\alpha) = \beta^T \hat{\mu} + \Phi^{-1}(\alpha)\sqrt{\beta^T \hat{\Sigma} \beta},$$

with $\Phi^{-1}(\alpha)$, as usual, the α quantile for the standard normal distribution.

The method just described is also known as the **variance-covariance method** and is very suitable for simulations as well. This method allows for including correlations among risk factors in the calculations. We stress the fact that this approach can be difficult to implement if we remove that assumption of linear pricing, but it would be still doable, albeit more complex.

In this day and age, simulating a multivariate normal in Python or most other packages is straightforward:

```
import numpy as np
# specify mean and variance-covariance matrix
mu = [1, 2]
Sigma = [[5, 1], [1, 4]]
```

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Simulating from other multivariate distributions can be more challenging. For the multivariate t-distribution in Python, one implementation is presented here: multivariate-t-distribution.

The R programming language has more built-in packages for this purpose, and if you are dealing with tasks related to VaR and ES, keep this in mind. Regardless, the multivariate t-distribution is quite important because it provides a good model for the returns of individual assets as seen in our earlier example. In addition, like the normal, the multivariate t-distribution has the property that the univariate marginal distributions have univariate t-distributions.

We can create a vector $Y \sim t(\nu, \mu, \Omega)$ by letting

$$Y = \mu + \sqrt{\frac{\nu}{V}} \cdot X \quad \text{where} \quad X \sim N(0, \Omega), V \sim \chi^2(\nu)$$

and V, X are independent.

Practitioners must note that Ω is the variance-covariance matrix for X, but not for Y The variance-covariance matrix for Y is

$$\text{var}(Y) = \frac{\nu}{\nu - 2}\Omega$$

and, as one can see, it is not even defined for $\nu <= 2$. However, Ω and Σ are just multiples of each other. If all the terms off the main diagonal of Ω are zero, then we would obtain vectors $m \times 1$ of independent univariate components.

In applications, there is still a need to estimate ν, μ, and Ω. This is usually done using MLE.

If we let Y describe the m individual assets in the portfolio and, for simplicity, we assume these are all stocks, then the linear pricing assumption is satisfied. Furthermore, in this case, the sensitivity factors β are all equal to 1.

In this scenario, if we start with $Y \sim t(\nu, \mu, \Omega)$ and calculate the estimates $\hat{\nu}, \hat{\mu}$, and $\hat{\Omega}$ then the portfolio, $\Pi = w^T Y$ has a univariate t-distribution with mean $w^T \hat{\nu}$ and variance $\frac{\hat{\nu}}{\hat{\nu}-2} w^T \hat{\Omega} w$ where w is the $m \times 1$ vector of the weights of each of the m-assets in the portfolio.

And here is some Python code to do just that:

```
# simulating from multivariate t distribution.
import numpy as np
from scipy.stats import chi

def multivariate_t_rvs(m, S, df=1, n=1):
m = np.asarray(m)  # procedure requires a type array
d = len(m)     # distribution is d-dimensional
v = np.random.chisquare(df, n)
z = np.random.multivariate_normal(np.zeros(d),S,(n,))
return m + z*np.sqrt(df/v)[:,None]

# using the function:
n=5  # number of simulations
df=3 # degrees of freedom
mu = [1, 2, -1]
Omega = [[1, .5, -0.3], [.5, 2, 0], [-0.3, 0, 2]]
simulated_values = multivariate_t_rvs(mu, Omega, df, n)
```

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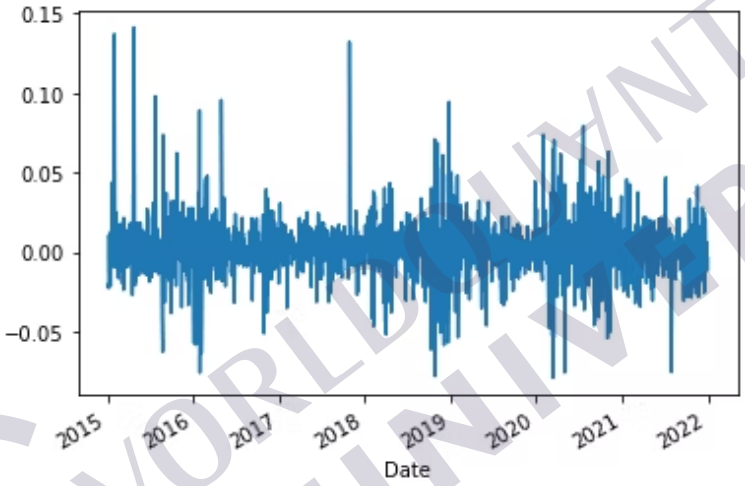
```
[[-0.15946801 -1.37601537 -8.26035747]
 [ 2.2442207   0.27457816 -2.11121623]
 [ 4.54699115  3.48056433 -3.79188743]
 [ 2.80460098  3.57288629 -4.10279591]
 [ 1.96680752  2.07122964 -2.89336293]]
```

This allows us to devise simulations where stock returns exhibit heavy tails, can be correlated among themselves, and will produce more accurate values of VaR and ES for our portfolios or investments.

4. VaR with Non-Stationary Distributions

In what we did so far, we made a very strong assumption: The distributions are stationary, meaning they do not change over time. However, we know that this is not the case in real life, not always at least since real data tend to exhibit a special type of heteroscedasticity known as volatility clustering.

Figure 4: Example of volatility clusters over different sub periods using returns on Amazon stock.



Therefore, one has to account for the time-varying volatility. Assume that we have n returns $r_1, r_2, \ldots, r_n$ and we need to estimate VaR and ES for the next return r_{n+1}. Let $\hat{\mu}_{n+1|r_1,\ldots,r_n}$ and $\hat{\sigma}_{n+1|r_1,\ldots,r_n}$ be the estimated and conditional variance of r_{n+1}, which are obtained conditionally on the current information available to us, i.e.$\{r_1, r_2, \ldots, r_n\}$. If we use a normal distribution, we could just compute VaR and ES using the same formula as before:

$$\mathbf{VaR}(1-\alpha) = \hat{\mu}_{n+1|r_1,\ldots,r_n} + \hat{\sigma}_{n+1|r_1,\ldots,r_n}\Phi^{-1}(\alpha)$$

and

$$\mathbf{ES}(1-\alpha) = \hat{\mu}_{n+1|r_1,\ldots,r_n} + \hat{\sigma}_{n+1|r_1,\ldots,r_n}\left(\frac{\phi(\Phi^{-1}(\alpha))}{\alpha}\right).$$

A similar results holds for the t-distribution if we

$$\widehat{\mathbf{VaR}}_p(1-\alpha) = M \cdot \left(\hat{\mu}_{n+1|r_1,\ldots,r_n} + \hat{\sigma}_{n+1|r_1,\ldots,r_n} \cdot F^{-1}_{\hat{\nu}_{n+1|r_1,\ldots,r_n}}(\alpha)\right)$$

and

$$\widehat{ES}_p(1-\alpha) = M \cdot \left(\hat{\mu}_{n+1|r_1,\ldots,r_n} - \hat{\sigma}_{n+1|r_1,\ldots,r_n} \times \left(\frac{f_{\hat{\nu}_{n+1|r_1,\ldots,r_n}}(F^{1-}_{\hat{\nu}_{n+1|r_1,\ldots,r_n}}(\alpha))}{\alpha}\right)\right)$$
$$\cdot \left[\frac{\hat{\nu}_{n+1|r_1,\ldots,r_n} + (F^{-1}_{\hat{\nu}_{n+1|r_1,\ldots,r_n}}(\alpha))^2}{\hat{\nu}_{n+1|r_1,\ldots,r_n} - 1}\right].$$

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that proposed by Riskmetrics, which uses an **exponentially weighted moving average** (*EWMA*). This method assumes that volatility, measured as σ_t^2 changes over time according to the following process:

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)(r_t - \mu)^2$$

with μ the unconditional mean and λ a constant arbitrarily chosen from the interval $(0, 1)$. Riskmetrics suggested $\lambda = 0.94.$ One should note that by the nature of how the exponential distribution decays, exponential weighting states that the older the history, the lower its effect.

If we compared this approach to using historical volatility, since the historical volatility using the most recent n observation is given by the formula, then:

$$\sigma_t^2 = \frac{1}{n}\sum_{i=t-n+1}^{t}(r_t - \mu)^2.$$

Then, simple algebraic manipulations yield

$$\sigma_t^2 = \frac{n-1}{n}\sigma_{t-1}^2 + \frac{1}{n}(r_n - \mu)^2.$$

In particular, we should notice that the historical volatility is a special case of EWMA and is achieved when $\lambda = (n-1)/n$. For n very large, then $(n-1)/n$ approaches 1 and the last observation receives less weight than choosing $\lambda = 0.94$ as proposed by Riskmetrics.

It would be preferable to estimate the value of λ directly from the data, and this leads us to use models like GARCH. In particular, we are going to illustrate this method using a GARCH(1, 1), a typical model for this type of application:

$$r_t = \mu + \epsilon_t, \;\; \epsilon_t \sim N(0, \sigma_t^2)$$
$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

This model can be implemented in Python using the following code:

```
from arch import arch_model

model=arch_model(stock_returns, vol='Garch', p=1, o=0, q=1, dist='Normal')
results=model.fit()
print(results.summary())
```

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```
Dep. Variable:                    stock   R-squared:                       0.000
Mean Model:             Constant Mean   Adj. R-squared:                  0.000
Vol Model:                      GARCH   Log-Likelihood:                4608.01
Distribution:                  Normal   AIC:                           -9208.02
Method:            Maximum Likelihood   BIC:                           -9186.12
                                        No. Observations:                 1762
Date:                Sun, Jan 09 2022   Df Residuals:                     1761
Time:                        09:29:48   Df Model:                            1
                            Mean Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
mu         2.2447e-03  4.750e-04      4.725  2.296e-06 [1.314e-03,3.176e-03]
                          Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega      3.5941e-05  3.846e-06      9.345  9.214e-21 [2.840e-05,4.348e-05]
alpha[1]       0.2000  3.749e-02      5.334  9.583e-08     [  0.127,  0.273]
beta[1]        0.7000  3.561e-02     19.655  5.241e-86     [  0.630,  0.770]
============================================================================

Covariance estimator: robust
```

Our estimated model is thus

$$\widehat{\sigma}_t^2 = 0.0000359 + 0.200\epsilon_{t-1}^2 + 0.700\sigma_{t-1}^2$$

and all estimates are statistically significant at the 95% confidence level. The model can now be used to predict the variance/volatility for our asset:

```
forecasts = results.forecast(horizon=3)
print(forecasts.variance)
h.1       h.2       h.3
Date
...       ...       ...
2021-12-27       NaN       NaN       NaN
2021-12-28       NaN       NaN       NaN
2021-12-29       NaN       NaN       NaN
2021-12-30       NaN       NaN       NaN
2021-12-31  0.000197  0.000213  0.000228
```

The 1-day estimated variance is 0.000197, which corresponds to a daily volatility of 0.0140, lower than the one we estimated using the historical values.

We can repeat the estimation using a t-distribution:

$$r_t = \mu + \epsilon_t, \;\; \epsilon_t \sim t_\nu$$
$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

```
model=arch_model(100*stock_returns, vol='Garch', p=1, o=0, q=1, dist='t')
```

Note that the multiplication by 100 was done to improve the chances of the estimating process converging to the proper solution and avoiding numerical inaccuracies. When estimating GARCH models, complex numerical procedures are

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The result is:

```
                     Constant Mean - GARCH Model Results
==============================================================================
Dep. Variable:                      stock   R-squared:                       0.000
Mean Model:                  Constant Mean   Adj. R-squared:                  0.000
Vol Model:                         GARCH   Log-Likelihood:               -3331.78
Distribution:      Standardized Student's t   AIC:                          6673.55
Method:                  Maximum Likelihood   BIC:                          6700.93
                                              No. Observations:                1762
Date:                      Sun, Jan 09 2022   Df Residuals:                    1761
Time:                             12:31:36   Df Model:                           1
                                Mean Model
============================================================================
                 coef    std err          t      P>|t|    95.0% Conf. Int.
----------------------------------------------------------------------------
mu             0.1540  3.118e-02      4.939  7.834e-07 [9.290e-02,  0.215]
                           Volatility Model
============================================================================
                 coef    std err          t      P>|t|    95.0% Conf. Int.
----------------------------------------------------------------------------
omega          0.2342  8.356e-02      2.803  5.063e-03 [7.044e-02,  0.398]
alpha[1]       0.1721  4.256e-02      4.044  5.260e-05 [8.869e-02,  0.256]
beta[1]        0.7787  5.081e-02     15.327  5.030e-53   [ 0.679,  0.878]
                            Distribution
============================================================================
                 coef    std err          t      P>|t|  95.0% Conf. Int.
----------------------------------------------------------------------------
nu             4.0709      0.435      9.365  7.585e-21 [  3.219,  4.923]
```

Note that, again, all estimates are statistically significant at the 95% confidence level and that in this case we have an additional estimate, $\hat{\nu}$, which is the estimate for the degrees of freedom of the t-distribution that is used in our GARCH(1,1) model. To obtain the forecast for the variance, we just need:

```
forecasts = results.forecast(horizon=3)
print(forecasts.variance)

h.1        h.2        h.3
Date
...        ...        ...
2021-12-27        NaN        NaN        NaN
2021-12-28        NaN        NaN        NaN
2021-12-29        NaN        NaN        NaN
2021-12-30        NaN        NaN        NaN
2021-12-31   1.988585   2.125057   2.254822
```

We must recall that we multiplied the stock returns by 100 for the sake of numerical accuracy in running the optimization algorithm, so we need to divide the number by 100^2 (this is the variance!). Hence, the 1-day estimate for the variance is $1.9886 \times 100^{-2} = 0.0001988$ or a volatility of 0.0141. Again, this is smaller than what we found using parametric or non-parametric estimation. It is often the case when using conditional approaches like GARCH.

If we are using a GARCH(1,1) model, we can also compute the unconditional variance using the formula

$$\hat{\sigma}^2_{uncond.} = \frac{\omega}{1-\alpha-\beta} = \frac{.2342}{1-.1721-.7787} = 4.7602$$

which must be divided by 100^2 to yield $\hat{\sigma}^2_{uncond.} = 0.000476$ for the data before the multiplication by 100. That translates to a daily volatility of 0.0218, considerably higher than the conditional one we estimated. The formula also shows that in order for the model to make sense, we need to check the condition that $\alpha + \beta < 1$.

Once the degrees of freedom for the t-distribution are known, we can use the formula we have seen earlier to compute $\mathbf{VaR}$ and $\mathbf{ES}$:

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and

$$\widehat{ES}_p(1-\alpha) = M \cdot \left(\hat{\mu} - \hat{\sigma} \left(\frac{f_{\hat{\nu}}(F_{\hat{\nu}}^{1-}(\alpha))}{\alpha} \right) \cdot \left[\frac{\hat{\nu} + (F_{\hat{\nu}}^{-1}(\alpha))^2}{\hat{\nu} - 1} \right] \right)$$

We only need to notice that while $\hat{\nu}$ can be taken from the output, $\hat{\lambda}_{n+1|r_1,\ldots,r_n}$ must be computed in the formula for λ and the estimate $\hat{\sigma}_{n+1|r_1,\ldots,r_n}$ that we just found, i.e., $\hat{\sigma}^2_{n+1|r_1,\ldots,r_n} = 0.0001988$.
Hence:

$$\hat{\lambda}_{n+1|r_1,\ldots,r_1} = \sqrt{\frac{\hat{\nu}-2}{\hat{\nu}}} \cdot \hat{\sigma}_{n+1|r_1,\ldots,r_n}.$$

Finally, $\hat{\mu}$ is the estimate "mu" in the output provided, i.e.$\hat{\mu}_{n+1|r_1,\ldots,r_n} = 0.1540/100 = .00154$ (recall we multiplied the returns by 100 to facilitate the convergence of the numerical computations).

One should notice that, in our example, the conditional volatility is below the unconditional, but it is not always so. As the graph shows, we are in a period of less turbulence for the stock, but at different times, we could find that the conditional variance is above the unconditional one. That is also in the spirit of volatility clustering.

This is not the "end of the story" because with time series models there are many other possible models that we could use. For instance, we could use a GARCH(1,2) (it was tried, and it does not produce a better statistical model), a combination of ARMA+GARCH, GARCH-M (GARCH-in-Mean), EGARCH (Exponential GARCH), TARCH (Threshold GARCH), and more. Of course, we could consider multivariate GARCH models as well.

5. Criticism of VaR

Despite the fact that VaR has become the major risk measure in the financial industry worldwide both among industry participants and regulators, VaR has also had its amount of detractors. Among the major shortcomings we find:

Liquidity risk. To understand this weakness, assume as it is often the case that we are working with a daily VaR. Implicit in the calculation of VaR is the assumption that positions can be completely liquidated very quickly, over a period of just a few days at most and at the prevailing market prices. However, during periods of particular economic turbulence, this may not be the case, and when major market participants starts to liquidate their positions, this could also cause a frenzy that induces other investors to do the same (herding behavior, one of the tenets of behavioral finance), and this determines a rapid drop in prices as everyone is trying to sell.

Model risk. We have seen that there are many ways to calculate VaR, but we need to assume some statistical model. The non-parametric approach does not; however, it is not good at estimating very small quantiles unless one has a lot of data. In other words, the non-parametric approach may be suitable for a95% $\mathbf{VaR}$, but not so for a99%-$\mathbf{VaR}$. Invalid assumptions can affect our calculations. In addition, a model may just look at past factors and fail to incorporate risk factors that are changing over time. This is also a risk that can occur when doing simulations as we are imposing certain distributions and certain relationships among the risk factors.

Quantiles of distributions do not tell us the whole story. As it is conceived, VaR only tells us that we have a probability α to lose more than a certain amount; however, it does not tell us the potential magnitude of those losses. Nevertheless, the expected shortfall allows us to say something about what happens on the negative tails of returns distribution, but it does not prevent institutions from taking very risky positions that could lead to significant losses, despite the low probability (assuming we are using the correct statistical model).

Lack of subadditivity. In an ideal world one expects that, because of diversification, the sum of the risks of two or more individual investments in a portfolio would not exceed the risk of the entire portfolio, i.e., if P_1 and P_2 represent two positions or portfolios, we expect that:

$$\mathbf{VaR}(P_1 + P_2) \leq \mathbf{VaR}(P_1) + \mathbf{VaR}(P_2).$$

However, this is not so with VaR, not always at least.

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of -c with probability 0.008. (We need a negative return, a loss, here for the example to work. Note: a return of -c is the same as saying a loss of c). The same is true for asset B. Then, $VaR_A(0.99) = VaR_B(0.99) = 0.$ Now, consider the portfolio A+B. In this case, we have a return of -2c with probability 0.008*0.008 = 0.000064, a return of - c with probability 2*0.008*0.992 = 0.01587 and a return of 0 with probability 0.98406. Then, $VaR_{A+B}(0.99) = -c.$ Hence, in this case, the VaR(0.99) for the portfolio is higher than the sum of the VaRs for the two assets that form the portfolio when considered as standalone assets. This is what is referred to as "lack of subadditivity.

The fact that the VaR of a sum may be higher than the sum of VaRs for each position a portfolio produces can cause financial institutions to create portfolios that tend to be concentrated. This can happen because a financial institution will tend to concentrate on large positions in a few assets, trying to game the VaR. In this scenario, given that the risk centers would be fewer, the VaR would be lower as well. Hence, the failure to properly measure the benefits of diversification is a significant issue.

Modeling individual assets together and also taking into consideration the correlations among the different asset classes does help to alleviate this issue. The following paper contains an interesting discussion on the limitations of diversification: "The Impact of Systemic Risk on the Diversification Benefits of a Risk Portfolio."

6. A Remark: From 1-day VaR to n-day VaR

In what we have done so far, we used the 1-day VaR, but not everyone does. As we discussed above, among the criticism of VaR is that in periods of downturn markets, it might be difficult to liquidate positions, in particular when dealing in sizable in illiquid assets. If it were to take several days to liquidate positions, a 10-day VaR could make more sense and provide a more realistic picture of the risk a financial institution faces. Here, there are two main approaches. For the sake of keeping things simple, we compare a 1-day VaR to a 30-day VaR. To obtain the 30-day VaR, we can follow two approaches:

We can take 30-day returns instead of 1-day returns. This is very simple and does not cause complications. However, we would be able to extract only 12 data points per year. That may not be enough to achieve the desired statistical precision in parametric approaches and definitely not enough to obtain reliable estimates using non-parametric methods, in particular on the tails of the distribution where precision requires a very large number of data points.

Alternatively, we could use a rolling average, i.e., we are considering periods of 30-days, but the periods are overlapping. This approach increases the number of data points with which we can work. However, now the 30-day returns are not uncorrelated as they use much of the same entries.

It is common practice to compute 1—day VaR and then scale it up to n-day using scaling factors: The mean used in computing the $30-day$ VaR is taken from that of the 1-day VaR and multiplied by 30; the standard deviation for the 30-day VaR is obtained by multiplying the volatility of the 1-day VaR by $\sqrt{30}$. However, this is correct only in those cases when prices follow a geometric Brownian motion and the returns a Brownian motion since in that case the properties of the normal distribution would allow us to do so.

7. Further Readings

An excellent paper that you should read at this point is "On Exactitude in Financial Regulation: Value-at-Risk, Expected Shortfall, and Expectiles." This paper summarizes several of the concepts we have seen in the lesson and expands on new ones, including the concept of *expectiles.* This is not a required reading, but it's highly encouraged.

8. Conclusion

In this lesson, we learned how to use VaR to assess portfolio risk and the severity of potential losses within given levels of confidence. This is very important. However, we need to also learn how to create good portfolios so that we can avoid risk as much as possible without giving up on investing in risky assets which could be necessary to generate certain target returns. This will be the scope of the next lesson.

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Lesson Notes

MODULE 1 | LESSON 3

FROM UTILITY THEORY TO CLASSICAL PORTFOLIO THEORY

Reading Time	120 minutes
Prior Knowledge	Matrices, algebra, some differential calculus
Keywords	Utility theory and relevant specific examples (CARA, CRRA, logarithmic utility, quadratic utility), The problem of maximizing utility and Markowitz portfolio theory, Quadratic programming

In the previous lesson, we looked at portfolio risk and how to measure the severity of possible losses using several approaches to VaR. In this lesson, we are looking at how to build optimal portfolios and how to incorporate certain constraints.

1. Elements of Utility Theory

The mean-variance model of asset choice has been used extensively in finance since Markowitz developed it back in 1952. To understand some of the nuances behind this model, one has to start with at least a modicum of utility theory.

There is clearly a difference between deciding which used car to buy, what brand of hamburger to eat, which project to choose for a company, or, generally, choosing among outcomes we do not know in advance. If a person decides to buy a share of Apple or Bitcoin, they do not know the outcome of their choice because the stock price follows a stochastic process. If one buys a meal, then we can assume that one knows.

There is also more than one way of not knowing the outcomes of certain events. For example, if we decide to play a roulette wheel, we do not know the outcome, but we could compute the probabilities for every outcome. However, we would not be able to do that in the case of betting on horse races. This has a consequence: Everyone who can perform the required calculations would agree on the likelihood of each outcome while playing a roulette wheel, but this is not so when betting on horse races. When the probabilities of outcomes are objective (i.e., different people w compute the same probability for an event), we have what we could describe as **risk**; on the

TOP

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Traditional portfolio theory assumes that all individuals are rational and capable of performing the required calculations to optimize their behavior and choices. Is that reasonable? Perhaps not, but it allows us to use higher mathematics to solve problems. Non-traditional approaches to portfolio theory will remove the assumption that all individuals have perfect information and/or optimize their every single decision. But that is a story for another part of the course.

2. Expected Values do not Work Well

When dealing with two or more risky situations, most of us, acting rationally, would find it sensible to choose the action that guarantees the highest average payoff, whenever the expected payoffs are greater than the cost of betting, of course. If it were not the case, then we would decide not to take the bet(s).

This is what one learns from early on dealing with the tools of probability and statistics. This approach is known as the **expected value approach**. However, this type of strategy can produce unforeseen consequences.

For instance, there is a famous example attributed to Nicolaus Bernoulli in 1713 and known as the **St. Petersburg paradox** , which we will cover in a moment, and a second one which shows some of the complications that ensue if we use an expected value approach, as well.

Example 1: (St. Petersburg paradox). Given a fair coin, we play the following game:

- **Step 1**. Flip a coin. If the outcome is heads (H), we win $\$2$;
- **Step 1**. If the outcome is tails (T), we flip the coin again.
- **Step 2**. If the coin comes up H, we win $\$2^2$;
- **Step 2**. If the coin comes up T, we flip the coin again: $\ldots$
- **Step n**. If the coin comes up H, we win $\$2^n$;
- **Step n**. If the coin comes up T, we flip the coin again: $\ldots$ forever.

The expected value for this game is then relatively easy to compute:

$$\text{expected value} = \sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} 1 = \infty.$$

How many times should we pay to play this game? Apparently, an infinite amount, i.e., one should play the game at any price, if offered the opportunity.

However, Daniel Bernoulli in his *Specimen Theoriae Novae de Mensura Sortis* (Exposition of a New Theory on the Measurement of Risk) in 1738 published in the *Commentarii Academiae Scientiarum Imperialis Petropolitanae* (Papers of the Imperial Academy of Sciences in Petersburg) reported that his cousin Nicolaus Bernoulli wrote in a letter to him that—after deriving this result and describing the game with an initial stake of 1 ducat—although the value of the expectation was infinitely great, "it [illegible] to be admitted that any fairly reasonable man would sell his chance, with great pleasure, for t[illegible] ducats."

TOP

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of the Econometric Society, pp. 23–36, 1954.

So, it seems that although one cannot lose at this game, 20 ducats may induce a person to choose this option, despite the lesser payoff. The second example is equally dramatic.

Example 2. (The pauper and the lottery). A poor person receives a magic lottery ticket, one that has a 50% chance to win \$1 million and a 50% chance to win \$0. A rich person offers to buy the ticket off the poor person for \$499,999 with certainty.

If the expected value approach held, the poor person would refuse the offer. However, most of us would agree that this would be unlikely as the difference between the expected value, \$500,000, and \$499,999 is minuscule, while the difference between \$0 and \$499,999 is massive for a poor person. The difference between \$1 million and \$499,999 is large but not as dramatic. However, choosing the \$499,999 invalidates the expected value approach, and yet that is what most people would do as shown in experiments.

The St. Petersburg paradox is so well-known that it has its own entries in encyclopedias and it can be found in discussions ranging from philosophy to probability to finance, etc. (for instance, Stanford Encyclopedia of Philosophy).

This paradox has faced several criticisms among which is the fact that "anyone who offers to let the agent play the St. Petersburg gamble is a liar, for he is pretending to have an indefinitely large bank" (Jeffrey), but it has also found those who defend it. Nevertheless, the second example is much less debatable and shows that the attitude towards wealth cannot be regarded just as an attitude towards an amount of money or monetary means and that we will have to consider it in terms of its utility to an economic agent.

To this purpose, casinos in Las Vegas fix a maximum bet of \$5,000 on roulette. Hence, if someone lost 12 times in a row, they would be unable to bet $\$2^{13} = \$8,192$. Consequently, the St. Petersburg gamble could not happen in Las Vegas.

3. Expected Utility Theory

Daniel Bernoulli suggested that people maximize expected utility, not expected value. To carry forward this program one has to introduce a **utility function**, say $U(w)$, and which represents the expected utility of an amount of wealth w. The function should be increasing ("more is better"), but the marginal utility should be decreasing (the value of an additional dollar value gets lower with each additional amount of wealth). Mathematically, this means that a utility function, $U(\cdot)$, should have the following properties:

$$dU(w)/dw > 0, \quad \text{and} \quad d^2U(w)/dw^2 < 0.$$

In particular, the negative second derivative alerts us to the fact that $U(\cdot)$ should be concave.

Example 3. (St. Petersburg paradox, continued). Let's assume that the player has the following utility function: $U(w) = \ln(w)$. It is easy to check that this utility function satisfies the two requirements we asked of a utility function. Then:

TOP

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The implications are even further reaching. In fact, we could consider the fact that if a person has an initial wealth of w_0,, then the value of the expected utility would be

$$E[U(w)] = \sum_{n=1}^{\infty} ln(2^n + w_0)\frac{1}{2^n} < \infty,$$

and it would be different, depending on how wealthy the gambler is originally. Thus, different agents would have different valuations for the same gambling games.

Unlike in the case of the expected value approach, Bernoulli is proposing that the majority of people do not extract utility from wealth per se, but rather, they extract utility from consuming goods, which is achieved by using that wealth to purchase the goods. This means that there is possibly a complicated nonlinear relationship between wealth and utility of wealth (that is, consumption). Nevertheless, Bernoulli tells us that this relationship is characterized by a utility function that is both increasing in wealth and concave (that is, it is increasing, but at a decreasing rate). Once we agree to adopt a utility theory approach, then we would choose x over y if $E[U(x)] > E[U(y)]$. We are still using the expected value, but in a very different fashion.

At this time, we assume that individuals are alive for only one period, which implies that individuals make decisions to use all their final wealth immediately to purchase/consume goods and services.

4. Risk-Averse Agents and Concave Utility Functions

We need to start with a formal definition, one that allows us to avoid running into troubles when using the everyday language instead.

Definition. An agent is **risk-averse** if, at any wealth level w, they dislike every lottery with an expected payoff of zero; i.e.,

$$E[U(w + z)] \leq E[U(w)]$$

for every lottery z such that $E[z] = 0$.

We should notice that every lottery, z, is a random variable and, as such, it can be written as $z = E[z] + (z - E[z])$; i.e., z can be decomposed into its expected payoff $E[z]$ and a zero mean lottery $z - E[z]$. Then, the definition of a risk-averse agent as given above can be reformulated by saying that:

A risk-averse agent always prefers receiving the expected outcome of a lottery with probability 1, rather than the lottery itself. For an expected utility maximizer, this implies that for any lottery z and for any initial wealth w :

$$E[U(w + z)] \leq U(w + E[z]).$$

The last relationship is always true for a concave function $U(\cdot)$ (Jensen's inequality for random variables), and therefore, we can infer that assuming a concave utility function is equivalent to saying that the economic agent is risk-averse. In fact, we have the following stronger result:

TOP

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Consider now a risk-averse agent, i.e., someone who dislikes zero-mean risk. That does not mean that they avoid any lottery like the plague. Instead, the agent may still like risky lotteries if the expected payoff is sufficiently large. More specifically, risk-averse investors may still want to buy risky assets if the expected return exceeds the cost to buy insurance.

This raises the following question: How is the optimal trade-off between the expected gain and the degrees of risk determined? Individuals are different: Some are willing to spend more than others to get rid of a specific risk. If π represents the risk premium, then for an agent with utility function $U(\cdot)$ and initial wealth w, the risk premium should satisfy the following equation:

$$E[U(w+z)] = U(w-\pi).$$

The agent ends up with the same amount of utility either by accepting the risk of the zero-mean lottery z or by paying the risk premium π to avoid the uncertainty of the lottery.

Since we know from earlier that, for a risk-averse agent, we have $E[U(w+z) \leq U(w+E[z])$, π must be nonnegative! The risk-premium is a complex function of the probability distribution of z and of the utility function $U(\cdot)$. Nevertheless, if we consider infinitesimal changes in the amounts of risk z, we could use a second-order Taylor approximation:

$$U(w-\pi) \approx U(w) - U'(w)\pi$$

and

$$\begin{aligned} E[U(w+z)] &\approx E[U(w) + U'(w)z + \frac{1}{2}U''(w)z^2] \\ &= U(w) + U'(w)E[z] + \frac{1}{2}U''(w)E[z^2] \\ &= U(w) + \frac{1}{2}U''(w)\sigma_z^2 \end{aligned}$$

where we used the fact that $E[z] = 0$ and let $\sigma_z^2 := E[z^2]$, the variance of the zero-mean lottery. Then, replacing what we just found in the earlier equation $E[U(w+z)] = U(w-\pi)$, we find that

$$\pi \approx \frac{1}{2}\sigma^2 \frac{-U''(w)}{U'(w)}.$$

Note that π is nonnegative since $U''(w) < 0$ and $U'(w) > 0$ because of the characteristics of $U(\cdot)$ for risk-averse investors. Incidentally, the function

$$A(w) := \frac{-U''(w)}{U'(w)}$$

which is called the **Arrow-Pratt measure of absolute risk-aversion** provides a good measure of risk aversion. The higher $A(w)$, the higher the risk aversion of the agent. We can see that this is the case because, everything else being equal, it tells us what amount an agent is willing to pay for insurance. The more risk-averse they are, the more they would pay to eliminate the risk.

A TOP r measure frequently used is the following:

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which is called a **measure of relative risk-aversion**. To understand the reason behind this name, one should notice that

$$\pi_r = \frac{\pi}{w} \approx -\frac{1}{2}\frac{U''(w)w}{U'(w)}\frac{\sigma_z^2}{w^2} = -\frac{1}{2}\frac{U''(w)w]}{U'(w)}\sigma_{z/w}^2 = \frac{1}{2}R(w)\sigma_{z/w}^2.$$

In the case of relative risk aversion, it is the variance of z/w and not the variance of just z that is relevant, hence the name of relative risk aversion.

5. Some Relevant Utility Functions

So far, our treatment has been very theoretical and to a large extent expected utility theory is. However, to become more acquainted with the concepts and also to gain some ability to read technical papers and literature, it is convenient to look at a few "practical" examples and learn what to look for.

Example 4. (CARA utility function). Suppose that the utility function is given by

$$U(w) = 1 - e^{-\alpha w}, \;\; \alpha > 0.$$

Then $U'(w) = \alpha e^{-\alpha w}$ and $U''(w) = -\alpha^2 e^{-\alpha w}$. Therefore

$$A(w) = \frac{-U''(w)}{U'(w)} = \alpha.$$

i.e., this utility function has a **constant absolute risk aversion**, and for that reason, it is known as a CARA utility function. To check on an important property of this utility function, let's assume that a certain payoff of d is preferred to a lottery z. This means that $U(d) \geq E[U(z)] = \sum_z U(z)Pr[z]$. Since the utility function is the CARA utility function, this implies that

$$1 - e^{-\alpha d} \geq \sum_z (1 - e^{\alpha z})Pr[z] \quad \text{or}$$

$$1 - e^{-\alpha d} \geq 1 - \sum_z e^{-\alpha z}Pr[z] \quad \text{or}$$

$$e^{-\alpha d} \leq \sum_z e^{-\alpha z}Pr[z].$$

Consider now another amount $d' > 0$. Then, from the last inequality, we find

$$e^{-\alpha d'}e^{-\alpha d} \leq e^{-\alpha d'}\sum_z e^{-\alpha z}Pr[z],$$

$$1 - e^{-\alpha d'}e^{-\alpha d} \geq 1 - e^{-\alpha d'}\sum_z e^{-\alpha z}Pr[z].$$

The last inequality is easily seen to be

$$U(d + d') \geq \sum_z U(z + d')Pr[z] = E[U(z + d')],$$

TOP

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or losing $75 is the same regardless of whether the initial wealth is, say $100, $100,000, or $ 1,000,000. Is this realistic? Probably not.

Example 5. (CRRA utility function.)

Let

$$U(w) = \frac{w^{1-\beta} - 1}{1-\beta}, \ \ \beta > 0.$$

In this case, one shows with a little bit of calculus and algebra that

$$R(w) := wA(w) = \beta,$$

i.e., the relative risk aversion is constant, or in plain English, proportional changes in prizes for the lottery do not affect the attitude towards risk. For this very reason, this utility function is known as CRRA (**constant relative risk aversion)**.

Example 6. (Logarithmic utility function). The log-utility function

$$U(w) = ln(w)$$

has the property that

$$A(w) = 1/w,$$

i.e., absolute risk aversion declines with the level of wealth. Instead, relative risk aversion is constant.

Example 7 (Quadratic utility function).
For a quadratic utility function, i.e.

$$U(w) = w - \gamma w^2, \ \ \gamma > 0,$$

(what happens if we used $+\gamma$ instead of $-\gamma$?) One finds that

$$A(w) = \frac{2\gamma}{1 - 2\gamma w}$$

and $A'(w) = \frac{4\gamma^2}{(1-2\gamma w)^2} > 0$, which implies that the absolute risk aversion is increasing with w, which makes the quadratic function suitable to model investors who reduce the nominal amount invested in risky assets as their wealth increases. This makes risky investments inferior goods, unfortunately!

6. Types of Investors and Their Utility Functions

Because this course is aimed at students of finance, it is worthwhile to conclude this section by looking at how to categorize the different types of agents.

TOP

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or taking their chances. On the other hand, a {\bf risk-seeking agent} is a decision-maker who prefers a risky payoff over a sure amount equal to its expected value.

All three types of agents have a utility function that is increasing, $U'(\cdot) > 0$. However, unlike in the case of the risk-averse agent, for the risk-neutral investors, it is $U''(\cdot) = 0$, i.e., the utility function is linear. On the other hand, for risk-seeking investors, $U''(\cdot) > 0$ or the utility function is convex.

If we look at the equation for π, we could also categorize the different types of agents by stating that risk-averse agents are willing to pay to eliminate risk while risk-seeking agents are willing to pay to take risk.

Traditional portfolio theory assumes that agents are risk-averse, but behavioral finance does not. And, in real life, we come across individuals who derive utility from participating in "lotteries," whatever lottery might mean to each of them.

There is another way to look at the three different types of economic agents. How do they behave when they need to invest?

- Risk-averse investors believe that "something is better than nothing" and like to play it safe. They tend to choose safer investments to place their assets. Of course, their payout still needs to meet the investors' demand for a return on investment.
- Risk-neutral investors simply select the investment with the highest expected return and do not take into account the risk of investments in their decision-making process.
- Finally, risk-seeking investors tend to invest in stocks with high betas, speculative investments, junk bonds. Sometimes, they are outright gamblers. They are willing to take on a considerable amount of risk as long as the potential return is worth the risk of their investment.

7. From Utility Theory to Classical Portfolio Theory

The mean variance model of asset choice introduced by H. Markowitz in 1952 has been already seen by students in previous classes. Essentially, we are told that, for instance, to find the minimum variance portfolio compatible with a given desired return, one should solve the following problem:

$$\begin{aligned} &\min_{w_1,\ldots,w_m} \sum_{i=1}^{m}\sum_{j=1}^{m} \sigma_{ij} w_i w_j \\ &\text{subject to } \sum_{i=1}^{m} w_i r_i = r_{min} \\ &\sum_{i=1}^{m} w_i = 1. \end{aligned}$$

We can generalize this problem by writing it in a different form that is more suitable for the developments that we are going to pursue in the next lectures. To this purpose, we write it as follows:

TOP

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$$\text{subject to } \sum_{i=1}^{m} w_i = 1.$$

In other words, instead of just finding the portfolio that has the lowest variance among those who yields a given return r_p, we just find the portfolio that maximizes the following expected utility function:

$$E[U(W)] = E[W] - \frac{\lambda}{2}\text{Var}[W]$$

where W denotes wealth. In the case of a portfolio investment $W = \sum_{i=1}^{m} w_i A_i$ where the A_i's are the assets that form the portfolio and the w_i's are the weights of those assets in the portfolio, respectively. We should notice that the optimization process will try to select weights that maximize return
and, at the same time, minimize variance. The presence of the $\frac{\lambda}{2}$ factor allows us to model different levels of risk aversion (the higher λ the more risk-averse the investor).

We have made a big deal about anchoring portfolio decisions to utility theory. So, it is legitimate to ask ourselves how we can do that. To this purpose, let's consider an investor who has an exponential utility; i.e.:

$$U(W) = -e^{-\lambda W}$$

where $\lambda > 0$ and W as usual denotes wealth. It is easy to check that $A(W) = \lambda$, i.e. the absolute risk aversion is constant and the larger
λ is, the more risk averse the agent is, which matches what we have in our optimization program. Now, we make a further assumption and assume that the wealth, W, is normally distributed with mean μ variance σ^2. In other words, we are assuming that the probability density function for W is

$$f(W) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(1/2\sigma^2)(W-\mu)^2}.$$

Then, the expected utility is given by

$$E[U(W)] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} -e^{-\lambda w} e^{-(1/2\sigma^2)(W-\mu)^2}\, dW$$

or,

$$E[U(W)] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} -e^{-(\lambda W + (1/2\sigma^2)(W-\mu)^2)}\, dW.$$

Now, a simple algebraic trick which is common among Statisticians and Probabilists when using normal random variables the way we are doing here
will prove very useful. More specifically, the exponent in the exponential function can be written in form:

TOP

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Using this equality in the previous formula, we get

$$E[U(W)] = \frac{e^{-\lambda(\mu-(1/2)\lambda\sigma^2)}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(W-\mu+\lambda\sigma^2)^2}{2\sigma^2}} dW.$$

At this point we note that the exponential inside the integral sign is just the pdf of a $N(\mu - \lambda\sigma^2, \sigma^2)$ random variable and thus

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(W-\mu+\lambda\sigma^2)^2}{2\sigma^2}} dW = 1.$$

Therefore we have just shown that

$$E[U(w)] = -e^{-\lambda(\mu-(1/2)\lambda\sigma^2)}$$

which does not even depend directly on W. Finally, if we want to maximize $E[U(w)]$, it is clear that we need maximize the expression

$$\mu - \frac{\lambda}{2}\sigma^2.$$

Considering that the mean for our portfolio is given by $\sum_{i=1}^{m} w_i r_i$ and the variance by $\sum_{i=1}^{m}\sum_{j=1}^{m} \sigma_{ij} w_i w_j$, we have shown
that the problem is indeed anchored in utility theory, at least as long as we use an exponential utility function as done above.

Remark 1. Someone may wonder whether if we could have found the same result using a quadratic utility function like $U(W) = \alpha W + \beta W^2$ and without making any assumption about the distribution of W. Well, not quite. In this case, we would find that

$$E[U(w)] = \alpha E[W] + \beta E[W^2] = \alpha E[W] + \beta[\mathbf{Var}[W] + (E[W])^2)$$

and we notice that we have an extra term, $(E[W])^2$, inside the square brackets on the right-hand side. However if, say $E[W] = 0.10$, then $(E[W])^2 = 0.01$ and therefore we could considered it negligible, but technically we do not get the same problem as written at the beginning of this section. Of course, it would be an equally easy problem to solve replacing the variance of the wealth with its second moment instead.

Remark 2. The result was obtained by assuming a specific utility function for our generic investor. What happens if we are not willing to assume that particular utility function? To investigate this point in a very general setup, let's assume that we have a utility function $U(W)$ whose functional form is arbitrary. Take a Taylor's series expansion of this function about $E[w]$. But, do this in this way:

$$U(W) = U(E[W)] + (W - E[W]))$$

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TOP

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where

$$R = \sum_{k=3}^{\infty} \frac{1}{k!} U^{(k)}(E[W])(W - E[W])^k.$$

In Markowitz setup, there is an assumption that investors decide about their investments based on the first two moments, i.e. $R = 0.$ Then, take expectation on both sides and we have

$$E[U(W)] = U(E[W]) + \frac{1}{2} U''(E[W]) Var[W].$$

If we assume that the typical investor is risk averse, then $U''(E[W]) < 0.$ This leads us to the classical MVO setup, but only if we are willing to assume that the extra terms (moments) in the Taylor's expansion can be set to zero or are at least negligible. Whether this is justified or not will depend on the true and usually unknown distribution of the wealth process $W.$ Notice for instance that if assumed again that W is normally distributed, then all odd central moments would be zero, but not the even moments. However, we could show that the even moments are related to the variance, but we will not pursue this here as it gets complicated and fast. Or we would need to assume a utility function whose derivatives past the second are null. The quadratic utility function seen above has this feature, but it does not cut the profile completely as we saw already.

Research has shown that if there is a riskless asset, then the distribution of every portfolio is determined by its mean and variance if and only if the random returns are a linear transformation of a spherically distributed random vector. This includes, among other distribution, the normal distribution. If there are no riskless assets, but returns are such that the last $n - 1$ components are spherically distributed conditional on the first component, which is allowed to have any distribution, then the distribution is also determined by its mean and variance. Students interested about this parts could read the following paper: G. Chamberlain, *A Characterization of the Distributions That Imply Mean-Variance Utility Functions*, Journal of Economic Theory, 29, p. 185-201 (1983) which contains the result mentioned above and a relatively rich list of references that discuss the issue going back to the 1960s.

8. Conclusion

In this lesson, we showed that economic agents are not driven simply by monetary amounts. The same monetary amount means different things to different people, depending on their initial wealth. This created the need to introduced utility theory to describe the actions and decisions of economic agents.

After introducing utility theory, we applied it to the classical portfolio optimization problem and showed that, in its standard formulation, such problem is oversimplified since quadratic utility functions are not completely satisfactory, but it has the merit to make the problem relatively simple to deal with in a mathematical rigorous manner.

In the next lesson, we will study how to solve the problem of finding optimal portfolios, determining t cient frontiers, identifying minimum variance and the tangency portfolios, and handling some

TOP

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Lesson Notes

MODULE 1 | LESSON 4

THE MATHEMATICS OF CLASSICAL PORTFOLIO THEORY

Reading Time	120 minutes
Prior Knowledge	None
Keywords	Portfolio frontier, Efficient frontier, Minimum global variance portfolio, Tangency portfolio, Language of quadratic programming, Formulating a mean-variance optimization problem in the language of QP.

The previous lesson showed us how to go from utility theory to propose a program to maximize an investor's utility through a mean-variance optimization problem. In this lesson, we introduce the mathematics required to solve the optimization problem and look at a few of the nuances of the topic.

1. Finding the Portfolio Efficient Frontier

Given a set of N assets, we choose to characterize a portfolio by the vector of its weights, i.e. $w^T = (w_1, w_2, \ldots, w_N)$. Recall that for us a vector is a column vector. Hence in our setup, w is an $N \times 1$ column vector, w^T, the transpose of w is a $1 \times N$ row vector, instead.

A portfolio is a frontier portfolio if it has the minimum variance among portfolios that have the same expected rate of return, that is if and only if the portfolio weights w_e^T are the solution of the following quadratic program:\

$$\begin{aligned} &\min_w \frac{1}{2} w^T \Sigma w \\ &\text{subject to } \; w^T r = E[r_p] \;\; \text{and} \\ &\qquad\qquad w^T \iota = 1, \end{aligned}$$

with Σ the $N \times N$ covariance matrix of the assets returns, r the $N \times 1$ vector of returns, ι, the $N \times 1$ vector of ones, i.e., $\iota^T = (1, 1, \ldots, 1)$, and $E[r_p]$ the expected rate of return on the portfolio. Putting a $\frac{1}{2}$ in front of the variance of the portfolio is only for mathematical convenience and does not alter the optimization.

While we will not insist on the full derivation of all formulae, it is useful for a student to see how certain problems are solved. A quadratic program like the one above is solved and which contains no inequality constraints, is still solvable using the traditional methods of Calculus, e.g., the method of the Lagrange multipliers. The Lagrangian is defined as

$$L(w, \lambda_1, \lambda_2) := \frac{1}{2} w^T \Sigma w + \lambda_1 (E[r_p] - w^T r) + \lambda_2 (1 - w^T \iota).$$

Now, the problem can be solved in the usual way, i.e. using multivariable calculus for $L(\cdot)$, and finding the max or min of the function L which is a function of $N + 2$ variables since w is a vector of N elements and then we have λ_1 and λ_2. The Lagrange equations, equivalent to the first order conditions for a max or min, can be written as

$$\frac{\partial L(w, \lambda_1, \lambda_2)}{\partial w_i} = 0, \;\; i = 1, 2, \ldots, N;$$

$$\text{and } \; \frac{\partial L(w, \lambda_1, \lambda_2)}{\partial \lambda_1} = 0, \frac{\partial L(w, \lambda_1, \lambda_2)}{\partial \lambda_2} = 0.$$

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when N is large, and this is where differential calculus for matrices comes in handy. We will not provide a full theory of this and will limit ourselves to illustrating the relevant formulas that are found useful.

2. A Basic Introduction to Differential Calculus in Matrix Form

This is a very brief section and will only touch on very basic facts needed for our development, but students who are interested in this topic should study it. Matrix differential calculus is also an important tool of advanced econometrics.

We can think of $w^T \Sigma w$ as a function from $R^N \mapsto R$ defined by $f : w \mapsto f(w) := w^T \Sigma w$.

If we take the first derivative with respect to the vector w, this amounts to doing the following:

$$\frac{\partial w^T \Sigma w}{\partial w} = \begin{pmatrix} \frac{\partial w^T \Sigma w}{\partial w_1} \\ \frac{\partial w^T \Sigma w}{\partial w_2} \\ \vdots \\ \frac{\partial w^T \Sigma w}{\partial w_N} \end{pmatrix} = 2\Sigma w.$$

The proof is not difficult, albeit tedious, but the result is very simple; the language of matrices is powerful. We also have two other multivariable terms to derive $w^T r$ and $w^T \iota$ and which are the same type of function of w. This time, we consider the function $g : R^N \mapsto R$ defined as $g(w) = w^T r = \sum_{i=1}^{N} w_i r_i$. Then

$$\frac{\partial w^T r}{\partial w} = \begin{pmatrix} \frac{\partial w^T r}{\partial w_1} \\ \frac{\partial w^T r}{\partial w_2} \\ \vdots \\ \frac{\partial w^T r}{\partial w_N} \end{pmatrix} = r.$$

Using exactly the same steps, then we also find that $\frac{\partial w^T \iota}{\partial w} = \iota$.

3. Back to the Efficient Frontier Task

Using the results just found, we can now derive the first order conditions for the Lagrangian L :

$$\frac{\partial L}{\partial w} = \Sigma w - \lambda_1 r - \lambda_2 \iota = \mathbf{0}$$

$$\frac{\partial L}{\partial \lambda_1} = E[r_p] - w^T r = 0$$

$$\frac{\partial L}{\partial \lambda_2} = 1 - w^T \iota = 0,$$

with $\mathbf{0}$ an $N \times 1$ vector of zeroes. A very simple and elegant finding. In particular, since Σ is a positive definite matrix (variance-covariance matrices are!), the first order conditions are not only necessary, but also sufficient to find the global minimum.

Now we need to solve the system of equations we derived. From the first equation, we get

$$w = \lambda_1 \Sigma^{-1} r + \lambda_2 \Sigma^{-1} \iota.$$

If we multiply both the right and the left sides of this equation by r^T and use the second equation from the first order conditions, we get

$$E[r_p] = r^T w = \lambda_1 r^T \Sigma^{-1} r + \lambda_2 r^T \Sigma^{-1} \iota.$$

Doing the same, but this time multiplying by ι^T and using the third equation from the first order conditions, yields

$$1 = \iota^T w = \lambda_1 \iota^T \Sigma^{-1} r + \lambda_2 \iota^T \Sigma^{-1} \iota.$$

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$$\lambda_1 \iota^T \Sigma^{-1} r + \lambda_2 \iota^T \Sigma^{-1} \iota = 1$$

which can be solved with respect to λ_1 and λ_2. The further (simple) algebraic manipulations will be spared and the solutions provided:

$$\lambda_1 = \frac{AE[r_p] - B}{D}$$
$$\lambda_2 = \frac{C - BE[r_p]}{D},$$

where

$$A = \iota^T \Sigma^{-1} \iota,\ B = \iota^T \Sigma^{-1} r = r^T \Sigma^{-1} \iota,\ C = r^T \Sigma^{-1} r,\ D = AC - B^2.$$

Finally, substituting these in the formula for w above, we get

$$w^\star = \frac{1}{D}(C\Sigma^{-1}\iota - B\Sigma^{-1} r) + \frac{1}{D}(A\Sigma^{-1} r - B\Sigma^{-1}\iota)E[r_p].$$

Notice that the optimal weight has the form:

$$w_p^\star = a + bE[r_p]$$

with

$$a = \frac{1}{D}(C\Sigma^{-1}\iota - B\Sigma^{-1} r), \quad \text{and} \quad b = \frac{1}{D}(A\Sigma^{-1} r - B\Sigma^{-1}\iota).$$

a and b are vectors, clearly. We use the subscript p to make the point that the optimal weights depend on the specific selected expected rate of return, $E[r_p]$.

We say that w_p^* is a **frontier portfolio** and the set of all frontier portfolios is called the **portfolio frontier**.

4. Two-Fund Separation

Let A and B be portfolios with expected return equal 0 and 1, respectively and let w_A and w_B be the weights that characterize these portfolios according to the formula we found in the previous section. Then, $w_A = a$ and $w_B = a + b$.

Now, we want a portfolio with expected return $E[r_q]$. Ideally that requires the weights $w = a + bE[r_q]$ which requires a lot of computations every time we change $E[r_q]$. However it is possible to obtain those weights without solving an optimization problem and just relying on portfolios A and B. More specifically, we invest $1 - E[r_q]$ in portfolio A and $E[r_q]$ in portfolio B. Note that this is legitimate as the sum of the weights of the new portfolio formed with A and B is equal to 1. Let's verify that this portfolio yields a return of $E[r_q]$. Let's call the newly formed portfolio P. Then

$$E[r_p] = (1 - E[r_q]) * r_A + E[r_q] \cdot r_B = (1 - E[r_q]) * 0 + E[r_q] \cdot 1 = E[r_q].$$

Recall that A is a portfolio with a return 0 and B a portfolio with a return of 1. So the portfolio does what we need it to do in terms of expected return.
The weights of the new portfolio can be found in terms of the weights of A and B as well:

$$w_p = w_A * (1 - E[r_q]) + w_B(E[r_q]) = a(1 - E[r_q]) + (a + b)E[r_q] = a + bE[r_q] = w_q,$$

just as we needed.

This fact allows us to state the following result:

Theorem (Two-fund separation). *Every portfolio on the portfolio frontier can be generated by any two distinct frontier portfolios not having the same expected return.*

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p_1 and portfolio p_2 whose weights are γ and $1-\gamma$. Then, using the earlier remark, we can write:

$$\begin{aligned}\gamma w_{p_1} + (1-\gamma) w_{p_2} &= \gamma(a + bE[r_{p_1}]) + (1-\gamma)(a + bE[r_{p_2}]) \\ &= a + b(\gamma E[r_{p_1}] + (1-\gamma)E[r_{p_2}]) = a + hE[r_q] = w_q.\end{aligned}$$

5. Global Minimum Variance Portfolio

Our next task is to identify the weights associated with the **global minimum variance portfolio**. This is actually simpler to do than computing the weights of a portfolio on the portfolio frontier since it is enough to solve the following program:

$$\min_w \frac{1}{2} w^T \Sigma w$$

$$\text{subject to } \; w^t \iota = 1,$$

It is basically the same problem as before with the condition $w^T r = E[r_p]$ removed. After some algebra (students are invited to perform the calculations as an exercise) one gets:

$$w_{GMV} = \frac{\Sigma^{-1}\iota}{\iota^T \Sigma^{-1} \iota}.$$

The expected return on the global minimum variance portfolio is then given by

$$E[r_{GMV}] = w_{GMV}^T \times r = \frac{\iota^T \Sigma^{-1} r}{\iota^T \Sigma^{-1} \iota} = \frac{B}{A}.$$

If we compare the solution for the generic portfolio on the portfolio frontier, we see that the variance for the portfolio with $E[r_p]$ is greater than that of the global minimum variance portfolio when $E[r_p] > \frac{B}{A}$. Also, if we look at our formula for λ_1, the last inequality implies that the Lagrange multiplier for the mean return constraint is positive. The set of minimum variance portfolios that satisfy this condition is called the **mean-variance efficient set**.

When $E[r_p] < \frac{B}{A}$, then $\lambda_1 < 0$ which implies that a lower mean return is consistent with a higher variance (the return is less than that of the global minimum variance portfolio and the variance is higher than that of the global minimum variance portfolio). Hence, no economic agent who likes mean and dislikes variance would want to hold this portfolio.

The variance of the global minimum variance portfolio can be calculated using the formula:

$$w_{GMV}^T \Sigma w_{GMV} = \frac{\iota^T \Sigma^{1-} \Sigma \Sigma^{-1} \iota}{(\iota^T \Sigma^{-1} \iota)^2} = \frac{\iota^T \Sigma^1 \iota}{(\iota^T \Sigma^{-1} \iota)^2} = \frac{1}{\iota^T \Sigma^{-1} \iota}.$$

Example (Naive diversification). What is the variance of a portfolio that assigns the same weight to each of the N assets? In this case:

$$w = \frac{1}{N}\iota.$$

and the variance of the portfolio is

$$w^T \Sigma w = \frac{1}{N^2} \iota^T \Sigma \iota.$$

If we assume that all assets have the same variance and correlation, then with a few algebraic manipulations:

$$\frac{1}{N^2}\iota^T \Sigma \iota = \frac{1}{N^2}\left(\sum_{i=1}^N \sigma^2 + \sum_{i=1}^N \sum_{j=1, j\neq i}^N \sigma^2 \rho\right) = \sigma^2 \rho + \frac{1}{N}\sigma^2(1-\rho).$$

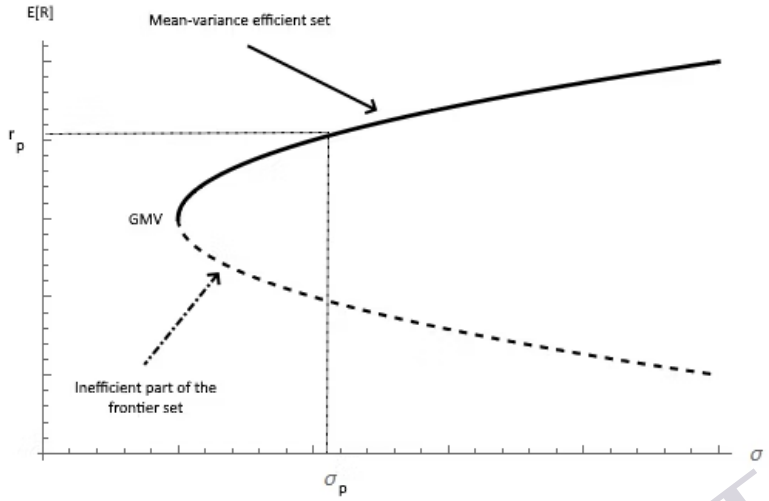
We know that the variance cannot be a negative number. This implies (the student is encouraged to do the algebra!) that

$$\rho > -\frac{1}{N-1}.$$

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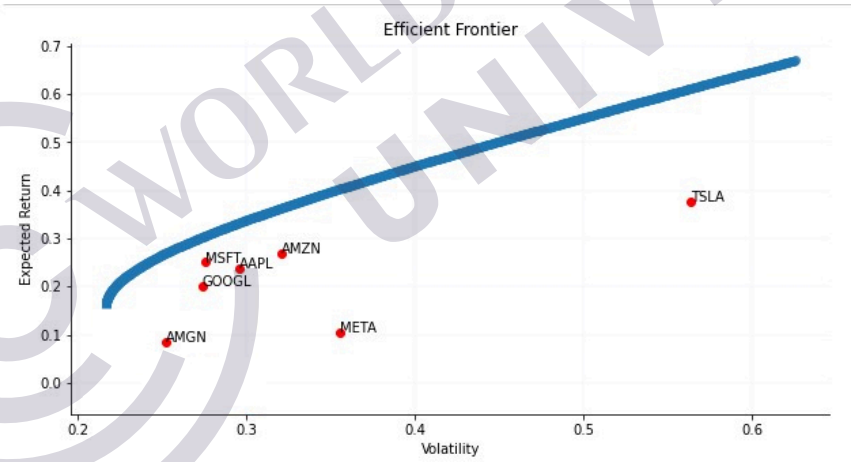
All our work can be summarized through the graph of Figure 1. Note that the dotted line represents the inefficient part of the frontier set.

Figure 1: The Mean-Variance Efficient Set and Global Minimum Variance Portfolio (GMV)



Example (Portfolio with 7 well known assets). We know enough to present a calculation with real assets. All the stock are well-known. We notice that at one side of the spectrum, Tesla has high return and high risk. On the other hand, Amgen, as a biotech/big pharma, has much less risk, but allows for a lower return. The efficient frontier for this example is exhibited in Figure 2.

Figure 2: A Computation of the Efficient Frontier Using a Group of Seven Assets (reference period Jan. 2, 2015–Jul. 7, 2022)



6. Adding a Riskless Asset and Tangency Portfolio

Until now, we considered all assets to be risky. Now we assume that there is a riskless asset. More specifically, we assume that we have N risky assets and 1 riskless asset. The risk-free rate is denoted by r_f. When a riskless asset is introduced, certain interesting developments take place.

Let p denote a frontier portfolio that consists of all $N+1$ assets, and use w_p for the $N \times 1$ vector of portfolio weights of p that relate just to the risky assets. Then, w_p can be obtained as the solution of the following quadratic program:

$$\min_{w_1, w_N} \frac{1}{2} w^T \Sigma w$$

$$\text{subject to } w^T r + (1 - w^T \iota) r_f = E[r_p].$$

Note that when $1 - w^T \iota < 0$ that is equivalent to borrowing money at the risk-free rate to buy more of the risky assets, and, as students already noticed, we remove the second constraint, i.e., the weights of the risky assets no longer need to

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$$L(w_1, \dots w_N, \lambda) = \frac{1}{2} w^T \Sigma w + \lambda \big(E[r_p] - w^T r - (1 - w^T \iota) r_f \big).$$

The first order conditions are

$$\frac{\partial L(w_1, \dots, w_N, \lambda)}{\partial w} = \Sigma w + \lambda(-r + r_f \iota) = 0$$
$$\frac{\partial L(w_1, \dots, w_N, \lambda)}{\partial \lambda} = E[r_p] - w^T r - (1 - w^T \iota) r_f = 0.$$

Therefore, if w_p, λ are the solutions of the quadratic programming, w_p must satisfy the following system of equations:

$$\Sigma w_p = \lambda (r - r_f \iota), \;\; r_f + w_p^T (r - r_f \iota) = E[r_p].$$

The system can be solved, and it yields

$$\lambda = \frac{E[r_p] - r_f}{E},$$
$$w_p = \lambda \Sigma^{-1} (r - r_f \iota) = \Sigma^{-1} (r - r_f \iota) \frac{E[r_p] - r_f}{E},$$

with $E = (r - r_f \iota)^T \Sigma^{-1} (r - r_f \iota)$. The scalar quantity E can be rewritten using our original quantities A, B, C, i.e.:

$$E = C - 2 B r_f + A r_f^2.$$

Since $B^2 - AC < 0$ and $A > 0$ (it is a positive definite quadratic form because Σ is positive definite and therefore so is Σ^{-1} is too), the parabola that is associated with E is always above the x-axis, and thus, E is always positive. Then, the variance of the minimized portfolio is

$$\sigma_p^2 = w_p^T \Sigma w_p = \lambda^2 (r - r_f \iota)^T \Sigma^{-1} \Sigma \Sigma^{-1} (r - r_f \iota) = \lambda^2 E$$

and, substituting the formula found above for λ, finally yields

$$\sigma_p^2 = \frac{(E[r_p] - r_f)^2}{E}.$$

This implies that $|E[r_p] - r_f| = \sigma_p \sqrt{E}$ and that

$$\sigma_p = \begin{cases} \frac{E[r_p] - r_f}{\sqrt{E}} & \text{if } E[r_p] \geq r_f; \\ -\frac{E[r_p] - r_f}{\sqrt{E}} & \text{if } E[r_p] < r_f. \end{cases}$$

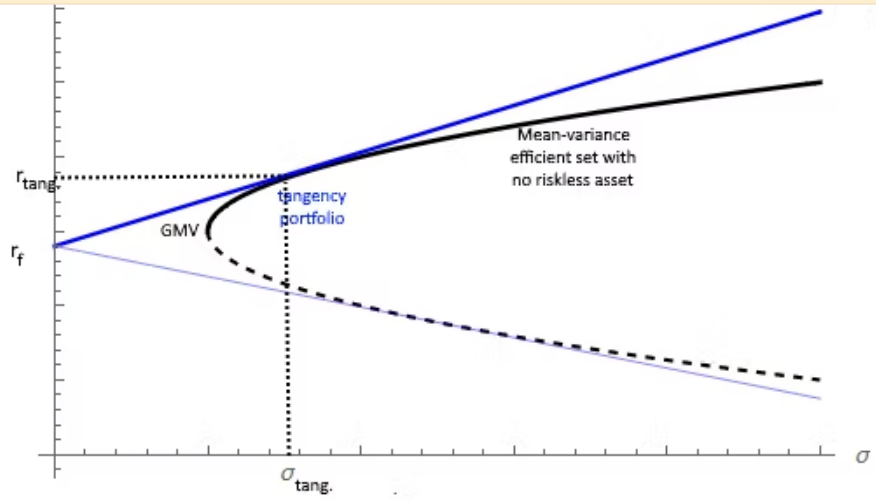
or, equivalently,

$$E[r_p] = \begin{cases} r_f + \sqrt{E} \sigma_p & \text{if } E[r_p] \geq r_f; \\ r_f - \sqrt{E} \sigma_p & \text{if } E[r_p] < r_f. \end{cases}$$

This result illustrates an important consequence: When we add a riskless asset and we allow borrowing funds at that rate, then the portfolio frontier becomes two half-lines that cross at the point $(0, r_f)$ in the $(\sigma, E[r])$ space.

Figure 3: Efficient Frontier and Tangency Portfolio with and without a Riskless Asset

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The upper branch of the V-shaped curve is a tangent line from r_f on the vertical axis to the mean-variance efficient set obtainable if one can only hold the risky assets and not the riskless asset. This upper branch is also known as the capital allocation line (CAL). Graphically, it appears as if there is only one portfolio on the CAL—and that contains only risky assets—and, as such, is also on the mean-variance efficient frontier constructed with risky assets only. This portfolio is known as the **tangency portfolio**. This can be observed in Figure 3.

One additional important feature of this portfolio is that the slope of such a portfolio is the same as that on the CAL line. We know that the slope of the CAL line is the Sharpe ratio.

There is one final question to address: how do we determine the tangency portfolio? The tangency portfolio is the solution of the following optimization problem:

$$\max_{w_1,\ldots,w_N} \frac{w^T r - r_f}{\sqrt{w^T \Sigma w}}$$
$$\text{subject to } w^T \iota = 1.$$

This is not an easy problem, although it can be solved using numerical algorithms.

If we assume that $w^T r - r_f > 0$, which is a reasonable assumption because all it says is that our portfolio is able to beat the risk-free rate, one can show (it is not straightforward, but not entirely difficult either) that the program above is equivalent to the following one:

$$\max_{v_1,\ldots,v_N} \frac{1}{\sqrt{v^T \Sigma v}}$$
$$\text{subject to } w^T \tilde{r} = 1.$$

with $\tilde{r} = r - r_f \iota$, or $\tilde{r}^T = (r_1 - r_f, r_2 - r_f, \ldots, r_N - r_f)$. Then, we can solve the problem to determine v as

$$\min_{v_1,\ldots,v_N} \frac{1}{2} v^T \Sigma v$$
$$\text{subject to } v^T \tilde{r} = 1,$$

which is a standard quadratic programming. Using the method of the Lagrange multipliers once again, one finds (students should do the calculations as in this case they are actually easy)

$$v = \frac{\Sigma^{-1} \tilde{r}}{\tilde{r}^T \Sigma^{-1} \tilde{r}}.$$

Once one has v, the weights for the tangency portfolio can be computed simply as

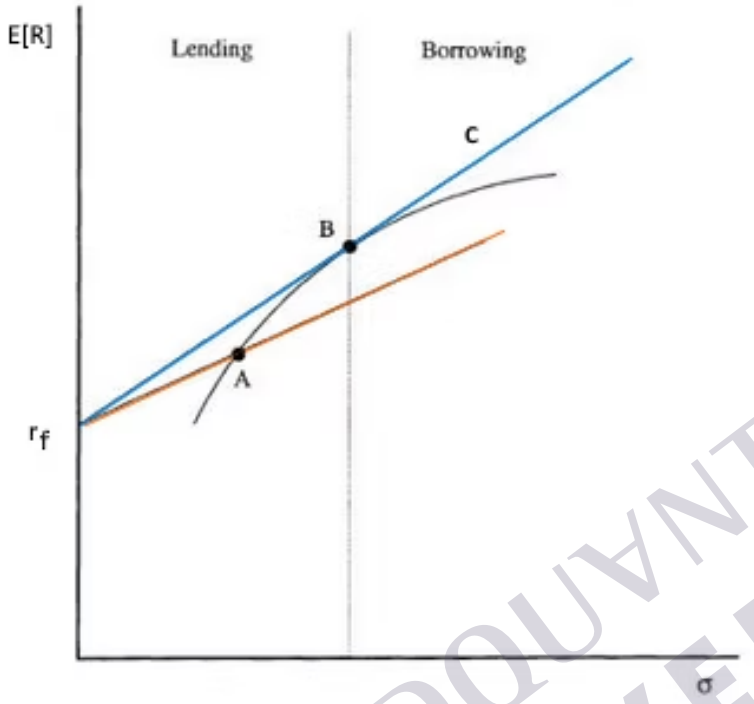
$$w_{TANG} = \frac{v}{v^T \iota} = \frac{\Sigma^{-1}(r - r_f \iota)}{B - r_f A}.$$

Where A and B are the same symbols used before.

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portfolio that contains only risky assets. In addition, this tells us one more thing: Different agents/investors may have a different appetite for risk (based on individual risk aversion); nevertheless, all investors should hold risky assets in the same proportion. The more conservative agents will dilute the risky assets by holding more of the risk-free asset, while aggressive agents may even borrow cash (at the risk-free rate) to increase their returns.

Figure 4: Different Portfolios for Different Investors



In figure 4, aggressive investors will actually borrow money and hold portfolio C. Investors who neither borrow nor lend money will hold the tangency portfolio B. The line that goes via portfolio A represents all combinations of risk-free lending and borrowing with portfolio A. These are not optimal combinations. An agent with zero appetite for risk will invest everything in a money market portfolio and earn the risk-free interest rate r_f.

Example (Putting the concepts to work). Assume that we have three risky assets whose expected returns are $0.05, 0.07$, and 0.11, respectively. Their individual volatilities are $0.10, 0.08$, and 0.15. The correlation between asset 1 and asset 2 is 0.4; the correlation between asset 1 and 3 is 0.5; and the correlation between asset 2 and 3 is -0.3. Finally, the risk-free interest rate is 0.01. What is the tangency portfolio?

First we need to compute the variance covariance matrix for these three assets. Students should check that it is:

$$\Sigma = \begin{pmatrix} 0.01 & 0.0032 & 0.0075 \\ 0.0032 & 0.0064 & -0.0036 \\ 0.0075 & -0.0036 & 0.0225 \end{pmatrix}$$

The vector $\tilde{r}^T = (r - r_f)^T = (0.04, 0.06, 0.10)$, instead. Then, we can find the weights for the portfolio v using the formula:

$$v = \frac{\Sigma^{-1}\tilde{r}}{\tilde{r}^T\Sigma^{-1}\tilde{r}}.$$

In order to do this we need first to invert the matrix Σ, which yields:

$$\Sigma^{-1} = \begin{pmatrix} 239.474 & -180.921 & -108.772 \\ -180.921 & 308.388 & 109.649 \\ -108.772 & 109.649 & 98.2456 \end{pmatrix}$$

and, after some algebra: $v^T = (-5.91986, 10.8287, 5.87073)$. Hence

$$w_{TANG}^T = (-0.549175, 1.00456, 0.544617).$$

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Let's say that an investor would like to achieve a return of 15% instead of just 10.65%. How can they achieve that? Clearly, they must be willing to accept more risk, but the mechanics are pretty simple and use a linear combination of the risk-free asset and the tangency portfolio; that is, we need to find α such that:

$$\alpha r_f + (1-\alpha) r_{TANG} = 0.15.$$

Using $r_f = 0.01$ and $r_{TANG} = 0.1028$, and solving the simple linear equation, we get:

$$\alpha = -0.5086,$$

i.e., we are borrowing at the risk-free rate (that we had to borrow it was clear since we are aiming for an expected return higher than that of the tangency portfolio) and using the additional funds to buy more of the tangent portfolio.

The weights for the portfolio consisting of the risk-free asset and the tangent portfolio are $(-0.5086, 1.5086)$.

To implement this portfolio, we need to specify the weights in terms of risk-free asset and the original three assets:

$$(-0.5086, 1.5086 \times (-0.549175, 1.00456, 0.544617)) = (-0.5086, -0.8285, 1.5155, 0.8216).$$

One can check that this portfolio allows us to achieve an expected return of 15% as desired. The volatility for the new portfolio is 9.79%, higher than that for the tangency portfolio.

7. Quadratic Programming (QP)

Classical portfolio theory lends itself to a certain amount of criticism, yet it is useful and addresses specific real life problems faced by practitioners in their profession. What was done in the previous sections is all good and elegant, but it is not enough to solve all problems.

In what we did so far, for instance, short-selling is allowed, we did not assign any limit to the weights of each individual stock in the portfolio, we did not consider the cost of rebalancing portfolios, etc. Whenever we introduce inequality constraints (like when we say that no individual stock can represent more than, say 10% of the overall portfolio, that is akin to stating that $w_i \leq 0.10$ for each $i = 1, 2, \ldots, N$. This introduces inequality constraints.

When inequality constraints are introduced, closed form solutions are no longer possible, not in general at least.

Thankfully, almost any high-level programming language makes available packages to perform quadratic programming.

In its most general form, a quadratic programming looks like this:

$$\begin{aligned} &\min_x \frac{1}{2} x^T P x + q^T x \\ &\text{subject to} \quad Gx = g \\ &\qquad\qquad\quad Hx \leq h \end{aligned}$$

where x is, say an $n \times 1$ vector with respect to that which we are trying to optimize.P is a positive definite matrix, q a vector, and G, H are suitable matrices to write the constraints. Note that the first set of constraints are equality constraints; the second set instead consists of inequality constraints. Furthermore, the inequality is to be understood as being applied element-wise to the vectors Hx and h.

If Python is the programming language of choice, the (CVXOPT) package contains a routine to do QP. So does CVXPY (CVXPY). There are more! Regardless of what software package is being used, one has to pay attention to whether the particular routine is using $\max$ or $\min$ since different packages may have different setups. However, this is not an issue at all since

$$\min_x \frac{1}{2} x^T P x + q^T x = \max_x -\left(\frac{1}{2} x^T P x + q^T x\right),$$

but, we stress it, one needs to be aware of how one's software of choice works.

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i.e., $w^T = (w_1, w_2, w_3), \iota^T = (1, 1, 1), r^T = (r_1, r_2, r_3)$.

The setup we used earlier was the following:

$$\begin{aligned} &\min_w \frac{1}{2} w^T \Sigma w \\ &\text{subject to } \; w^T r = E[r_p] \;\; \text{and} \\ &\qquad\qquad w^T \iota = 1. \end{aligned}$$

The first thing that we notice is that here we have no inequality constraints. Hence:

$$P = \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 \end{pmatrix}, \;\; q = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \;\; G = \begin{pmatrix} r_1 & r_2 & r_3 \\ 1 & 1 & 1 \end{pmatrix}, \;\; g = \begin{pmatrix} E[r_p] \\ 1 \end{pmatrix}$$

and we do not need any H matrix and h vector. However, some software (like the one used in the programming language R) write the matrices H and G as a single matrix and then one needs to tell the program how many rows pertain to equality constraints while the remaining ones are taken as inequality constraints.

Example (Portfolio optimization with limited max exposure to individual stocks). We can consider the same problem as above, i.e., short selling is allowed but we do not want the weight of any stock to exceed 0.40. Then, the problem becomes: stocks, i.e., $w^T = (w_1, w_2, w_3), \iota^T = (1, 1, 1), r^T = (r_1, r_2, r_3)$. The, setup we used earlier was the following:

$$\begin{aligned} &\min_w \tfrac{1}{2} w^T \Sigma w \\ &\text{subject to } \; w^T r = E[r_p] \;\; \text{and} \\ &\qquad\qquad w^T \iota = 1 \\ &\qquad\qquad w_1 \leq 0.4 \\ &\qquad\qquad w_2 \leq 0.4 \\ &\qquad\qquad w_3 \leq 0.4. \end{aligned}$$

In this case, the matrices P, G are as before and so are the vectors q and g. However, now we have:

$$H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \;\; h = \begin{pmatrix} 0.4 \\ 0.4 \\ 0.4 \end{pmatrix}.$$

Some clients have strong environmental, social, and governance (ESG) feelings and may require that their portfolio managers do not invest in certain classes of assets. In that case, weights for assets that are forbidden by clients would be set to zero.

Example (Portfolio optimization with limited max and min exposure to individual stocks (box constraints)). We can consider the same problem as above, i.e., short selling is allowed, but we do not want the weight of any stock to exceed 0.40 and, in addition, the client has requested that the weight of the third asset does not go below 0.2 Then, the problem becomes:

$$\begin{aligned} &\min_w \tfrac{1}{2} w^T \Sigma w \\ &\text{subject to } \; w^T r = E[r_p] \;\; \text{and} \\ &\qquad\qquad w^T \iota = 1 \\ &\qquad\qquad w_1 \leq 0.4 \\ &\qquad\qquad w_2 \leq 0.4 \\ &\qquad\qquad w_3 \leq 0.4 \\ &\qquad\qquad w_3 \geq 0.2. \end{aligned}$$

The last inequality can be written as $\geq$ by simply using it as $-w_3 \leq -0.2$. Hence, the matrices P, q, G, g remain as usual, but

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$$\begin{pmatrix} 0 & 0 & 1 & 0 & -1 \end{pmatrix} \qquad \begin{pmatrix} \\ -0.2 \end{pmatrix}$$

Now we look at a more concrete example with specific names from everyday life.

Example (Application to the portfolio consisting of AAPL, MSFT, AMZN, GOOGL, META, TSLA, AMGN). We have already seen this setup in deriving the portfolio frontier. Let's say that we want to find the optimal portfolio that yields 35 return with no short selling constraints, but the sum of the weights needs to be 1. Then (using daily data from Jan. 2, 2015, to Jan. 15, 2022 to estimate returns and parameters), the optimal solution is given by the following weights:

```
0.22251740

0.34298749

0.44328915

0.13602145

-0.16611600

0.11255463

-0.09125412
```

and the portfolio achieves a volatility of 0.2695.

Let's say now that we want the same return, but we do not want to be more than 0.05 in TSLA. Then, running a new quadratic program with an added constraint, we get

```
0.2537489

0.3877040

0.4809173

0.1310326

-0.1730621

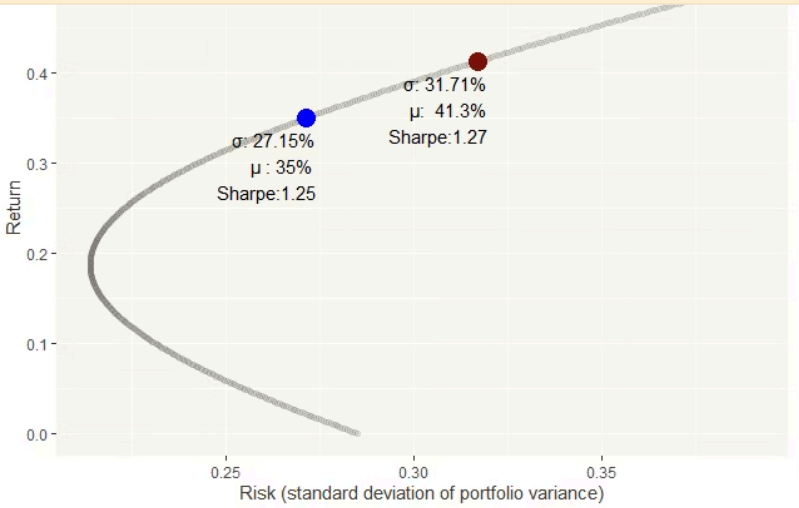
0.0500000

-0.1303407
```

The computed volatility is now 0.2715, higher than before and which was to be expected as we added an additional constraint. The Sharpe ratio for this portfolio is 1.25 (see figure 5).

Figure 5: Required Solution (blue dot) vs. the Best Portfolio (red dot)

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Looking at the portfolio frontier can be sort of fun, and it can look appealing, but ultimately, what really matters is the composition of the portfolio we need to implement for an institution or for individual clients. Still, one can produce plots like figure 5 by running the program for multiple values of desired expected returns.

To complete this section, we look at an interesting problem. What happens if lending and borrowing rates are not the same? Assume that r_l is the risk-free lending rate, and r_b is the borrowing rate. Then, we can define a new interest rate that depends on the particular situation, i.e., it is a function of the weights of the risky assets in the portfolio.

$$r(w) = \begin{cases} r_l & \text{if } w^T\iota - 1 \leq 0; \\ r_b & \text{if } w^T\iota - 1 \geq 0. \end{cases}$$

Using $r(w)$, the expected return for the portfolio can be written as:

$$E[r_p] = w^T r + (1 - w^t\iota)r(w).$$

and the portfolio frontier can be found by solving the programming problem:

$$\min_{w_1,\ldots,w_N} \frac{1}{2} w^T \Sigma w$$

$$\text{subject to } \; w^T r + (1 - w^T\iota)r(w) = E[r_p].$$

This is NOT a quadratic program as the function $r(w)$ is not smooth (it is not differentiable everywhere).

To solve this problem, one has to actually solve two problems:

1. If $w^T\iota > 1$:

$$\min_{w_1,\ldots,w_N} \frac{1}{2} w^T \Sigma w$$

$$\text{subject to } \; w^T r + (1 - w^T\iota)r_b = E[r_p].$$

1. and if $w^T\iota \leq 1$, then:

$$\min_{w_1,\ldots,w_N} \frac{1}{2} w^T \Sigma w$$

$$\text{subject to } \; w^T r + (1 - w^T\iota)r_l = E[r_p].$$

Once the two quadratic programs are solved, we obtain two systems of weights. The solution (portfolio) with the lower variance is the solution we want. QP is used for each of the two sub-problems though.

We conclude this section with some Python code where the CVXPY package is used to implement quadratic programming and determine the weights for an optimal portfolio when both equality and inequality constraints are used. Generally speaking, from a didactic point of view, it is much better when students write their own code. But as this skill is an

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now how to use pip install if your Anaconda is missing certain libraries.

```
import pandas as pd
import numpy as np
import cvxpy as cvx
import yfinance as yf
from datetime import datetime

## define the stocks for our portfolio (via tickers) ###
assets = ["AAPL", "AMZN","TSLA","GOOGL","META","MSFT","AMGN", "V"]
# Define the start date
start = "2020-03-15"
# Today's date
today = datetime.today().strftime('%Y-%m-%d')
# Acquire data
df_prices = yf.download(assets,start=start_date,end=today)['Adj Close']

# We transform the data to its logarithmic returns
df = np.log(df_prices).diff()
# Drop the first row because we loose information we the logarithmic return.
df = df.dropna()  # this is a dataframe

# Compute the Expected Returns, we multiply daily return by 252
#because there are 252 business days in the US.
exp_returns = df.mean()*252
cova = df.cov()*252

m=exp_returns.shape[0]

# WANT: optimal portfolio with the following constraints
# 1. no short sales are allowed  that is the weights for all stocks are >=0
# 2. the maximum weight allowed for Amazon is 0.2
# 3. the maxium weight allowed for Tesla is 0.3
# 4. the sum of the weights is 1
# 5. We need to have a weight of 0.05 for Microsoft

#def optimize_portfolio(exp_returns, index_weights, scale=.0000):
Q = np.array(cova)   # Q is variance-covariance matrix of assets
q = np.array(exp_returns) # q is the exptected returns of assets
#   Inequalities are written as Gx <= h
G = np.array(np.matrix([[-1,0,0,0,0,0,0,0],[0,-1,0,0,0,0,0,0], \
                        [0,0,-1,0,0,0,0,0],[0,0,0,-1,0,0,0,0], \
                        [0,0,0,0,-1,0,0,0],[0,0,0,0,0,-1,0,0], \
                        [0,0,0,0,0,0,-1,0],[0,0,0,0,0,0,0,-1], \
                        [0,1,0,0,0,0,0,0],[0,0,1,0,0,0,0,0]]))
h =np.array([0,0,0,0,0,0,0,0,0.2,0.3])
```

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```
                # matrix A is used for the equality constraints
b = np.array([1, 0.05])
#
# Define and solve the quadratic problem using CVXPY:
# ===================================================
x = cvx.Variable(m)
prob = cvx.Problem(cvx.Maximize((-1/2)*cvx.quad_form(x, Q) + q.T @ x),
            [G @ x <= h,
             A @ x == b])
prob.solve()
#retrieve the weights of the optimized portfolio
x_values = x.value
#Print the optimal weights for each asset in the portfolio.
for i in range(m):
        print(assets[i], f"{x_values[i]: f}")

opt_ptfolio_ret = np.dot(x_values, exp_returns)
print("Expected return optimal porfolio: ", f"{opt_ptfolio_ret:.2%}")

opt_ptfolio_stdev = np.sqrt(np.dot(x_values, Q.dot(x_values.T)))
print("Expected std. deviation optimal porfolio: ", f"{opt_ptfolio_stdev:.2%}")
```

8. Conclusion

In this lesson, we learned about the nuances of classical mean-variance optimization to determine optimal portfolios. We covered the math necessary to derive the important formula, and we looked at a real-life application.

Behind the scenes, estimates for the mean of expected returns for individual assets and their estimated covariance are fed to an algorithm. In the next lesson, we will investigate more in depth the issues related to using estimates. We anticipate that it is not a small role, unfortunately. In particular, it is the estimated mean returns that can cause the most trouble. In an attempt to mitigate the issue, we will visit resampling techniques using both a simple bootstrap technique and the celebrated Michaud's resampling approach.